GenIz

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2.1 Package List

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Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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5.1 File List

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- directory request
- directory response

6.2 src/main/java/capstoneProject Directory Reference

Directories

- directory Lorenzo

6.3 src/test/java/capstoneProject Directory Reference

Directories

- directory Lorenzo

6.4 src/main/java/capstoneProject/Lorenzo/genIz/controller Directory Reference

Files

- file DataBaseController.java
- file ExportDataController.java
- file GenerateQuizController.java
- file OnlineModelController.java
- file RESTController.java

6.5 src/main/java/capstoneProject/Lorenzo/genlz/DAO Directory Reference

Files

- file QuizDaoImpl.java
- file QuizDaoInterface.java

6.6 src/main/java/capstoneProject/Lorenzo/genlz/DTO Directory Reference

Directories

- directory entity_dto
- directory request_dto
- directory response_dto

Files

- file ErrorResponseDto.java
- file ResponseDataDto.java

6.7 src/main/java/capstoneProject/Lorenzo/genlz/entity Directory Reference

Files

- file ChatEntity.java
- file DiscussionEntity.java
- file OnlineModelEntity.java
- file UserEntity.java

6.8 src/main/java/capstoneProject/Lorenzo/genlz/DTO/entity_dto Directory Reference

Files

- file ChatDataDto.java
- file DiscussionDataDto.java
- file UserDataDTO.java

6.9 src/main/java/capstoneProject/Lorenzo/genlz Directory Reference

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- directory api_format
- directory controller
- directory DAO
- directory DTO
- directory entity
- directory global_exception_handler
- directory security
- directory service

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- file GenIzApplication.java

6.10 src/test/java/capstoneProject/Lorenzo/genlz Directory Reference

Files

- file GenIzApplicationTests.java
- file JsonValidationTest.java
- file UpdateOnlineModelSettingsTest.java

6.11 src/main/java/capstoneProject/Lorenzo/genlz/global_exception_handler Directory Reference

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- file GlobalExceptionHandler.java

6.12 src/main/java Directory Reference

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- directory capstoneProject

6.13 src/test/java Directory Reference

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- directory capstoneProject

6.14 src/main/java/capstoneProject/Lorenzo Directory Reference

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- directory genIz

6.15 src/test/java/capstoneProject/Lorenzo Directory Reference

Directories

- directory genIz

6.16 src/main Directory Reference

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- directory java

6.17 src/main/java/capstoneProject/Lorenzo/genIz/api_format/↩ response/map_model_response Directory Reference

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- file Choices.java
- file Message.java
- file TotalTokens.java

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- directory request_parameters

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- file ApiCallRequest.java

6.19 src/main/java/capstoneProject/Lorenzo/genlz/DTO/request_dto Directory Reference

Files

- file PostReqChatDto.java
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- file PostReqOnlineModelDto.java
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6.20 src/main/java/capstoneProject/Lorenzo/genlz/api_format/request/request_parameters Directory Reference

Files

- file QuizGenParameter.java

6.21 src/main/java/capstoneProject/Lorenzo/genlz/api_format/response Directory Reference

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- directory map_model_response

Files

- file ApiCallResponse.java

6.22 src/main/java/capstoneProject/Lorenzo/genlz/DTO/response_dto Directory Reference

Files

- file ResponseChatListDto.java
- file ResponseDiscussionListDto.java
- file ResponseModelSettingDto.java
- file ResponseOnlineModelDto.java
- file ResponseUserInfoDto.java

6.23 src/main/java/capstoneProject/Lorenzo/genlz/security Directory Reference

Files

- file SecurityConfig.java

6.24 src/main/java/capstoneProject/Lorenzo/genlz/service Directory Reference

Files

- file ApiFlowAggregatorInterface.java
- file ExportQuizToLocalMachine.java
- file ExportTxtFile.java
- file GenerateQuizApi.java
- file GenerateQuizApiInterface.java
- file ImplementQuizDaoMethodsInterface.java
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- file LocalModelApiFlowAggregator.java
- file QuizDaoService.java

6.25 src Directory Reference

Directories

- directory main
- directory test

6.26 src/test Directory Reference

Directories

- directory java

Chapter 7

Namespace Documentation

7.1 Package capstoneProject.Lorenzo.genlz

Packages

- package controller
- package DAO
- package DTO
- package entity
- package global_exception_handler
- package security
- package service

Classes

- class GenIzApplication
- class GenIzApplicationTests
- class JsonValidationTest
- class UpdateOnlineModelSettingsTest

7.2 Package capstoneProject.Lorenzo.genlz.api_format.request

Packages

- package request_parameters

Classes

- class ApiCallRequest

7.3 Package capstoneProject.Lorenzo.genlz.api_format.request.↩ request_parameters

Classes

- class QuizGenParameter

7.4 Package capstoneProject.Lorenzo.genlz.api_format.response

Packages

- package map_model_response

Classes

- class ApiCallResponse

7.5 Package capstoneProject.Lorenzo.genlz.api_format.response.map↩ _model_response

Classes

- class Choices
- class Message
- class TotalTokens

7.6 Package capstoneProject.Lorenzo.genlz.controller

Classes

- class DataBaseController
- class ExportDataController
- class GenerateQuizController
- class OnlineModelController
- class RESTController

7.7 Package capstoneProject.Lorenzo.genlz.DAO

Classes

- class QuizDaoImpl
- interface QuizDaoInterface

7.8 Package capstoneProject.Lorenzo.genlz.DTO

Packages

- package entity_dto
- package request_dto
- package response_dto

Classes

- class ErrorResponseDto
- class ResponseDataDto

7.9 Package capstoneProject.Lorenzo.genlz.DTO.entity_dto

Classes

- class ChatDataDto
- class DiscussionDataDto
- class UserDataDTO

7.10 Package capstoneProject.Lorenzo.genlz.DTO.request_dto

Classes

- class PostReqChatDto
- class PostReqDiscussionDto
- class PostReqModelSetting
- class PostReqOnlineModelDto
- class PostReqUserTextDto

7.11 Package capstoneProject.Lorenzo.genlz.DTO.response_dto

Classes

- class ResponseChatListDto
- class ResponseDiscussionListDto
- class ResponseModelSettingDto
- class ResponseOnlineModelDto
- class ResponseUserInfoDto

7.12 Package capstoneProject.Lorenzo.genlz.entity

Classes

- class ChatEntity
- class DiscussionEntity
- class OnlineModelEntity
- class UserEntity

7.13 Package capstoneProject.Lorenzo.genlz.global_exception_handler

Classes

- class GlobalExceptionHandler

7.14 Package capstoneProject.Lorenzo.genlz.security

Classes

- class SecurityConfig

7.15 Package capstoneProject.Lorenzo.genlz.service

Classes

- interface ApiFlowAggregatorInterface
- interface ExportQuizToLocalMachine
- class ExportTxtFile
- class GenerateQuizApi
- interface GenerateQuizApiInterface
- interface ImplementQuizDaoMethodsInterface
- class JsonValidation
- interface JsonValidationInterface
- class LocalModelApiFlowAggregator
- class QuizDaoService

Chapter 8

Class Documentation

8.1 capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest Class Reference

Public Member Functions

- ApiCallRequest ()
- String getModel ()
- void setModel (String model)
- List< Map< String, String > > getMessages ()
- void setMessages (List< Map< String, String > > messages)
- void addField (String role, String content)

Private Attributes

- String model
- List< Map< String, String > > messages = new ArrayList<>()

8.1.1 Detailed Description

Definition at line 8 of file ApiCallRequest.java.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 ApiCallRequest()

```
capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.ApiCallRequest ()
```

Definition at line 13 of file ApiCallRequest.java.

8.1.3 Member Function Documentation

8.1.3.1 addField()

```
void capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.addField (
            String role,
            String content)
```

Definition at line 35 of file ApiCallRequest.java.

8.1.3.2 getMessages()

```
List< Map< String, String > > capstoneProject.Lorenzo.genIz.api_format.request.ApiCall↩
Request.getMessages ()
```

Definition at line 26 of file ApiCallRequest.java.

8.1.3.3 getModel()

```
String capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.getModel ()
```

Definition at line 18 of file ApiCallRequest.java.

8.1.3.4 setMessages()

```
void capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.setMessages (
            List< Map< String, String > > messages)
```

Definition at line 30 of file ApiCallRequest.java.

8.1.3.5 setModel()

```
void capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.setModel (
            String model)
```

Definition at line 22 of file ApiCallRequest.java.

8.1.4 Member Data Documentation

8.1.4.1 messages

```
List<Map<String, String> > capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.↩
messages = new ArrayList<>()  [private]
```

Definition at line 10 of file ApiCallRequest.java.

8.1.4.2 model

```
String capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest.model  [private]
```

Definition at line 9 of file ApiCallRequest.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/api_format/request/ApiCallRequest.java

8.2 capstoneProject.Lorenzo.genIz.api_format.response.ApiCall↩ Response Class Reference

Public Member Functions

- List< Choices > getChoices ()
- void setChoices (List< Choices > choices)
- TotalTokens getUsage ()
- void setUsage (TotalTokens usage)

Private Attributes

- List< Choices > choices
- TotalTokens usage

8.2.1 Detailed Description

Definition at line 8 of file ApiCallResponse.java.

8.2.2 Member Function Documentation

8.2.2.1 getChoices()

```
List< Choices > capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse.getChoices
()
```

Definition at line 12 of file ApiCallResponse.java.

8.2.2.2 getUsage()

```
TotalTokens capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse.getUsage ()
```

Definition at line 18 of file ApiCallResponse.java.

8.2.2.3 setChoices()

```
void capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse.setChoices (
            List< Choices > choices)
```

Definition at line 15 of file ApiCallResponse.java.

8.2.2.4 setUsage()

```
void capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse.setUsage (
            TotalTokens usage)
```

Definition at line 21 of file ApiCallResponse.java.

8.2.3 Member Data Documentation

8.2.3.1 choices

```
List<Choices> capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse.choices  [private]
```

Definition at line 9 of file ApiCallResponse.java.

8.2.3.2 usage

```
TotalTokens capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse.usage  [private]
```

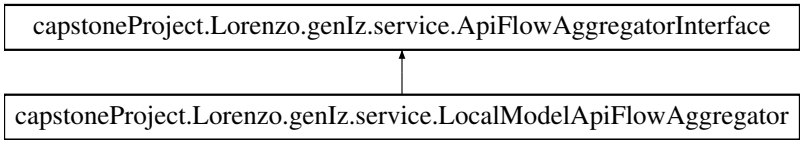
Definition at line 10 of file ApiCallResponse.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/ApiCallResponse.java

8.3 capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface Interface Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface:



Public Member Functions

- ResponseDataDto performApiCall (PostReqUserTextDto postReqUserTextDto)
- ResponseDataDto validateApiCallResult (ResponseDataDto tempDTO)

8.3.1 Detailed Description

Definition at line 7 of file ApiFlowAggregatorInterface.java.

8.3.2 Member Function Documentation

8.3.2.1 performApiCall()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface.performApi↩
Call (
            PostReqUserTextDto postReqUserTextDto)
```

Implemented in capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.

8.3.2.2 validateApiCallResult()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface.validateApi↩
CallResult (
            ResponseDataDto tempDTO)
```

Implemented in capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.

The documentation for this interface was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/ApiFlowAggregatorInterface.java

8.4 capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto Class Reference

Public Member Functions

- Integer getChat_id ()
- void setChat_id (Integer chat_id)
- String getChat_name ()
- void setChat_name (String chat_name)
- Integer getUser_id ()
- void setUser_id (Integer user_id)

Private Attributes

- Integer chat_id
- String chat_name
- Integer user_id

8.4.1 Detailed Description

Definition at line 3 of file ChatDataDto.java.

8.4.2 Member Function Documentation

8.4.2.1 getChat_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.getChat_id ()
```

Definition at line 8 of file ChatDataDto.java.

8.4.2.2 getChat_name()

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.getChat_name ()
```

Definition at line 14 of file ChatDataDto.java.

8.4.2.3 getUser_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.getUser_id ()
```

Definition at line 20 of file ChatDataDto.java.

8.4.2.4 setChat_id()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.setChat_id (
            Integer chat_id)
```

Definition at line 11 of file ChatDataDto.java.

8.4.2.5 setChat_name()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.setChat_name (
            String chat_name)
```

Definition at line 17 of file ChatDataDto.java.

8.4.2.6 setUser_id()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.setUser_id (
            Integer user_id)
```

Definition at line 23 of file ChatDataDto.java.

8.4.3 Member Data Documentation

8.4.3.1 chat_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.chat_id  [private]
```

Definition at line 4 of file ChatDataDto.java.

8.4.3.2 chat_name

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.chat_name  [private]
```

Definition at line 5 of file ChatDataDto.java.

8.4.3.3 user_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto.user_id  [private]
```

Definition at line 6 of file ChatDataDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/entity_dto/ChatDataDto.java

8.5 capstoneProject.Lorenzo.genIz.entity.ChatEntity Class Reference

Public Member Functions

- Integer getChat_id ()
- void setChat_id (Integer chat_id)
- String getChat_name ()
- void setChat_name (String chat_name)
- UserEntity getDefUserEntity ()
- void setDefUserEntity (UserEntity defUserEntity)
- List< DiscussionEntity > getDiscussionEntity ()
- void setDiscussionEntity (List< DiscussionEntity > discussionEntity)
- void addDiscussion (DiscussionEntity discussionEntity)

Private Attributes

- Integer chat_id
- String chat_name
- UserEntity defUserEntity
- List< DiscussionEntity > discussionEntityList

8.5.1 Detailed Description

Definition at line 19 of file ChatEntity.java.

8.5.2 Member Function Documentation

8.5.2.1 addDiscussion()

```
void capstoneProject.Lorenzo.genIz.entity.ChatEntity.addDiscussion (
            DiscussionEntity discussionEntity)
```

Definition at line 73 of file ChatEntity.java.

8.5.2.2 getChat_id()

```
Integer capstoneProject.Lorenzo.genIz.entity.ChatEntity.getChat_id ()
```

Definition at line 40 of file ChatEntity.java.

8.5.2.3 getChat_name()

```
String capstoneProject.Lorenzo.genIz.entity.ChatEntity.getChat_name ()
```

Definition at line 48 of file ChatEntity.java.

8.5.2.4 getDefUserEntity()

```
UserEntity capstoneProject.Lorenzo.genIz.entity.ChatEntity.getDefUserEntity ()
```

Definition at line 56 of file ChatEntity.java.

8.5.2.5 getDiscussionEntity()

```
List< DiscussionEntity > capstoneProject.Lorenzo.genIz.entity.ChatEntity.getDiscussionEntity
()
```

Definition at line 64 of file ChatEntity.java.

8.5.2.6 setChat_id()

```
void capstoneProject.Lorenzo.genIz.entity.ChatEntity.setChat_id (
            Integer chat_id)
```

Definition at line 44 of file ChatEntity.java.

8.5.2.7 setChat_name()

```
void capstoneProject.Lorenzo.genIz.entity.ChatEntity.setChat_name (
            String chat_name)
```

Definition at line 52 of file ChatEntity.java.

8.5.2.8 setDefUserEntity()

```
void capstoneProject.Lorenzo.genIz.entity.ChatEntity.setDefUserEntity (
            UserEntity defUserEntity)
```

Definition at line 60 of file ChatEntity.java.

8.5.2.9 setDiscussionEntity()

```
void capstoneProject.Lorenzo.genIz.entity.ChatEntity.setDiscussionEntity (
            List< DiscussionEntity > discussionEntity)
```

Definition at line 68 of file ChatEntity.java.

8.5.3 Member Data Documentation

8.5.3.1 chat_id

```
Integer capstoneProject.Lorenzo.genIz.entity.ChatEntity.chat_id  [private]
```

Definition at line 24 of file ChatEntity.java.

8.5.3.2 chat_name

```
String capstoneProject.Lorenzo.genIz.entity.ChatEntity.chat_name  [private]
```

Definition at line 27 of file ChatEntity.java.

8.5.3.3 defUserEntity

```
UserEntity capstoneProject.Lorenzo.genIz.entity.ChatEntity.defUserEntity  [private]
```

Definition at line 33 of file ChatEntity.java.

8.5.3.4 discussionEntityList

```
List<DiscussionEntity> capstoneProject.Lorenzo.genIz.entity.ChatEntity.discussionEntityList
[private]
```

Definition at line 38 of file ChatEntity.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/entity/ChatEntity.java

8.6 capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Choices Class Reference

Public Member Functions

- Message getMessage ()
- void setMessage (Message message)

Private Attributes

- Message message

8.6.1 Detailed Description

Definition at line 3 of file Choices.java.

8.6.2 Member Function Documentation

8.6.2.1 getMessage()

```
Message capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Choices.get↩
Message ()
```

Definition at line 6 of file Choices.java.

8.6.2.2 setMessage()

```
void capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Choices.setMessage (
            Message message)
```

Definition at line 10 of file Choices.java.

8.6.3 Member Data Documentation

8.6.3.1 message

```
Message capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Choices.message
[private]
```

Definition at line 4 of file Choices.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/map_model_response/Choices.java

8.7 capstoneProject.Lorenzo.genIz.controller.DataBaseController Class Reference

Public Member Functions

- DataBaseController (ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface)
- ResponseChatListDto saveChat (@RequestBody PostReqChatDto request)
- List< ResponseChatListDto > getAllChats ()
- ResponseDiscussionListDto saveDiscussion (@RequestBody PostReqDiscussionDto discussion)
- List< ResponseDiscussionListDto > getAllDiscussions (@PathVariable Integer chatId)
- ResponseModelSettingDto saveModelSetting (@RequestBody PostReqModelSetting modelSetting)
- ResponseModelSettingDto getModelSetting ()
- void deleteChat (@PathVariable Integer chatId)
- ResponseChatListDto patchChat (@RequestBody PostReqChatDto postReqChatDto)
- void deleteDiscussion (@PathVariable Integer discussionId)

Package Attributes

- ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface

8.7.1 Detailed Description

Definition at line 27 of file DataBaseController.java.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 DataBaseController()

```
capstoneProject.Lorenzo.genIz.controller.DataBaseController.DataBaseController (
            ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface)
```

Definition at line 32 of file DataBaseController.java.

8.7.3 Member Function Documentation

8.7.3.1 deleteChat()

```
void capstoneProject.Lorenzo.genIz.controller.DataBaseController.deleteChat (
            @PathVariable Integer chatId)
```

Definition at line 92 of file DataBaseController.java.

8.7.3.2 deleteDiscussion()

```
void capstoneProject.Lorenzo.genIz.controller.DataBaseController.deleteDiscussion (
            @PathVariable Integer discussionId)
```

Definition at line 108 of file DataBaseController.java.

8.7.3.3 getAllChats()

```
List< ResponseChatListDto > capstoneProject.Lorenzo.genIz.controller.DataBaseController.get↩
AllChats ()
```

Definition at line 47 of file DataBaseController.java.

8.7.3.4 getAllDiscussions()

```
List< ResponseDiscussionListDto > capstoneProject.Lorenzo.genIz.controller.DataBaseController.↩
getAllDiscussions (
            @PathVariable Integer chatId)
```

Definition at line 65 of file DataBaseController.java.

8.7.3.5 getModelSetting()

```
ResponseModelSettingDto capstoneProject.Lorenzo.genIz.controller.DataBaseController.getModel↩
Setting ()
```

Definition at line 83 of file DataBaseController.java.

8.7.3.6 patchChat()

```
ResponseChatListDto capstoneProject.Lorenzo.genIz.controller.DataBaseController.patchChat (
            @RequestBody PostReqChatDto postReqChatDto)
```

Definition at line 99 of file DataBaseController.java.

8.7.3.7 saveChat()

```
ResponseChatListDto capstoneProject.Lorenzo.genIz.controller.DataBaseController.saveChat (
            @RequestBody PostReqChatDto request)
```

Definition at line 38 of file DataBaseController.java.

8.7.3.8 saveDiscussion()

```
ResponseDiscussionListDto capstoneProject.Lorenzo.genIz.controller.DataBaseController.save↩
Discussion (
            @RequestBody PostReqDiscussionDto discussion)
```

Definition at line 56 of file DataBaseController.java.

8.7.3.9 saveModelSetting()

```
ResponseModelSettingDto capstoneProject.Lorenzo.genIz.controller.DataBaseController.save↩
ModelSetting (
            @RequestBody PostReqModelSetting modelSetting)
```

Definition at line 74 of file DataBaseController.java.

8.7.4 Member Data Documentation

8.7.4.1 implementQuizDaoMethodsInterface

```
ImplementQuizDaoMethodsInterface capstoneProject.Lorenzo.genIz.controller.DataBaseController.↩
implementQuizDaoMethodsInterface  [package]
```

Definition at line 30 of file DataBaseController.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/controller/DataBaseController.java

8.8 capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto Class Reference

Public Member Functions

- Integer getDiscussion_id ()
- void setDiscussion_id (Integer discussion_id)
- String getUser_pdf_name ()
- void setUser_pdf_name (String user_pdf_name)
- String getQuiz_content ()
- void setQuiz_content (String quiz_content)
- Integer getChat_id ()
- void setChat_id (Integer chat_id)

Private Attributes

- Integer discussion_id
- String user_pdf_name
- String quiz_content
- Integer chat_id

8.8.1 Detailed Description

Definition at line 3 of file DiscussionDataDto.java.

8.8.2 Member Function Documentation

8.8.2.1 getChat_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.getChat_id ()
```

Definition at line 27 of file DiscussionDataDto.java.

8.8.2.2 getDiscussion_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.getDiscussion_id ()
```

Definition at line 9 of file DiscussionDataDto.java.

8.8.2.3 getQuiz_content()

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.getQuiz_content ()
```

Definition at line 21 of file DiscussionDataDto.java.

8.8.2.4 getUser_pdf_name()

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.getUser_pdf_name ()
```

Definition at line 15 of file DiscussionDataDto.java.

8.8.2.5 setChat_id()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.setChat_id (
            Integer chat_id)
```

Definition at line 30 of file DiscussionDataDto.java.

8.8.2.6 setDiscussion_id()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.setDiscussion_id (
            Integer discussion_id)
```

Definition at line 12 of file DiscussionDataDto.java.

8.8.2.7 setQuiz_content()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.setQuiz_content (
            String quiz_content)
```

Definition at line 24 of file DiscussionDataDto.java.

8.8.2.8 setUser_pdf_name()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.setUser_pdf_name (
            String user_pdf_name)
```

Definition at line 18 of file DiscussionDataDto.java.

8.8.3 Member Data Documentation

8.8.3.1 chat_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.chat_id  [private]
```

Definition at line 7 of file DiscussionDataDto.java.

8.8.3.2 discussion_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.discussion_id  [private]
```

Definition at line 4 of file DiscussionDataDto.java.

8.8.3.3 quiz_content

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.quiz_content [private]
```

Definition at line 6 of file DiscussionDataDto.java.

8.8.3.4 user_pdf_name

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto.user_pdf_name [private]
```

Definition at line 5 of file DiscussionDataDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/entity_dto/DiscussionDataDto.java

8.9 capstoneProject.Lorenzo.genIz.entity.DiscussionEntity Class Reference

Public Member Functions

- Integer getDiscussion_id ()
- void setDiscussion_id (Integer discussion_id)
- String getUser_pdf_name ()
- void setUser_pdf_name (String user_pdf_name)
- String getQuiz_content ()
- void setQuiz_content (String quiz_content)
- ChatEntity getDefChatEntity ()
- void setDefChatEntity (ChatEntity defChatEntity)

Private Attributes

- Integer discussion_id
- String user_pdf_name
- String quiz_content
- ChatEntity defChatEntity

8.9.1 Detailed Description

Definition at line 14 of file DiscussionEntity.java.

8.9.2 Member Function Documentation

8.9.2.1 getDefChatEntity()

```
ChatEntity capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.getDefChatEntity ()
```

Definition at line 57 of file DiscussionEntity.java.

8.9.2.2 getDiscussion_id()

```
Integer capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.getDiscussion_id ()
```

Definition at line 33 of file DiscussionEntity.java.

8.9.2.3 getQuiz_content()

```
String capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.getQuiz_content ()
```

Definition at line 49 of file DiscussionEntity.java.

8.9.2.4 getUser_pdf_name()

```
String capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.getUser_pdf_name ()
```

Definition at line 41 of file DiscussionEntity.java.

8.9.2.5 setDefChatEntity()

```
void capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.setDefChatEntity (
            ChatEntity defChatEntity)
```

Definition at line 61 of file DiscussionEntity.java.

8.9.2.6 setDiscussion_id()

```
void capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.setDiscussion_id (
            Integer discussion_id)
```

Definition at line 37 of file DiscussionEntity.java.

8.9.2.7 setQuiz_content()

```
void capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.setQuiz_content (
            String quiz_content)
```

Definition at line 53 of file DiscussionEntity.java.

8.9.2.8 setUser_pdf_name()

```
void capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.setUser_pdf_name (
            String user_pdf_name)
```

Definition at line 45 of file DiscussionEntity.java.

8.9.3 Member Data Documentation

8.9.3.1 defChatEntity

```
ChatEntity capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.defChatEntity [private]
```

Definition at line 31 of file DiscussionEntity.java.

8.9.3.2 discussion_id

```
Integer capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.discussion_id [private]
```

Definition at line 19 of file DiscussionEntity.java.

8.9.3.3 quiz_content

```
String capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.quiz_content [private]
```

Definition at line 25 of file DiscussionEntity.java.

8.9.3.4 user_pdf_name

```
String capstoneProject.Lorenzo.genIz.entity.DiscussionEntity.user_pdf_name [private]
```

Definition at line 22 of file DiscussionEntity.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/entity/DiscussionEntity.java

8.10 capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto Class Reference

Public Member Functions

- ErrorResponseDto (String exceptionErrorMessage)
- String getExceptionErrorMessage ()
- void setExceptionErrorMessage (String exceptionErrorMessage)

Public Attributes

- String exceptionErrorMessage

8.10.1 Detailed Description

Definition at line 3 of file ErrorResponseDto.java.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 ErrorResponseDto()

```
capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto.ErrorResponseDto (
            String exceptionErrorMessage)
```

Definition at line 6 of file ErrorResponseDto.java.

8.10.3 Member Function Documentation

8.10.3.1 getExceptionErrorMessage()

```
String capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto.getExceptionErrorMessage ()
```

Definition at line 10 of file ErrorResponseDto.java.

8.10.3.2 setExceptionErrorMessage()

```
void capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto.setExceptionErrorMessage (
            String exceptionErrorMessage)
```

Definition at line 14 of file ErrorResponseDto.java.

8.10.4 Member Data Documentation

8.10.4.1 exceptionErrorMessage

```
String capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto.exceptionErrorMessage
```

Definition at line 4 of file ErrorResponseDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/ErrorResponseDto.java

8.11 capstoneProject.Lorenzo.genIz.controller.ExportDataController Class Reference

Public Member Functions

- ExportDataController (ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface, ExportQuizToLocalMachine exportQuizToLocalMachine)
- void postMethodName (@PathVariable Integer chatId)

Package Attributes

- ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface
- ExportQuizToLocalMachine exportQuizToLocalMachine

8.11.1 Detailed Description

Definition at line 18 of file ExportDataController.java.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 ExportDataController()

```
capstoneProject.Lorenzo.genIz.controller.ExportDataController.ExportDataController (
            ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface,
            ExportQuizToLocalMachine exportQuizToLocalMachine)
```

Definition at line 24 of file ExportDataController.java.

8.11.3 Member Function Documentation

8.11.3.1 postMethodName()

```
void capstoneProject.Lorenzo.genIz.controller.ExportDataController.postMethodName (
            @PathVariable Integer chatId)
```

Definition at line 32 of file ExportDataController.java.

8.11.4 Member Data Documentation

8.11.4.1 exportQuizToLocalMachine

```
ExportQuizToLocalMachine capstoneProject.Lorenzo.genIz.controller.ExportDataController.↩
exportQuizToLocalMachine  [package]
```

Definition at line 22 of file ExportDataController.java.

8.11.4.2 implementQuizDaoMethodsInterface

```
ImplementQuizDaoMethodsInterface capstoneProject.Lorenzo.genIz.controller.ExportDataController.↩
implementQuizDaoMethodsInterface  [package]
```

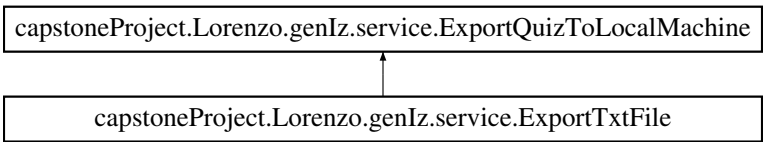
Definition at line 21 of file ExportDataController.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/controller/ExportDataController.java

8.12 capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine Interface Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine:



Public Member Functions

- List< String > formatFileContent (List< ResponseDiscussionListDto > quizList)
- void saveFile (List< String > fileContent)

8.12.1 Detailed Description

Definition at line 7 of file ExportQuizToLocalMachine.java.

8.12.2 Member Function Documentation

8.12.2.1 formatFileContent()

```
List< String > capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine.formatFile↩
Content (
            List< ResponseDiscussionListDto > quizList)
```

Implemented in capstoneProject.Lorenzo.genIz.service.ExportTxtFile.

8.12.2.2 saveFile()

```
void capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine.saveFile (
            List< String > fileContent)
```

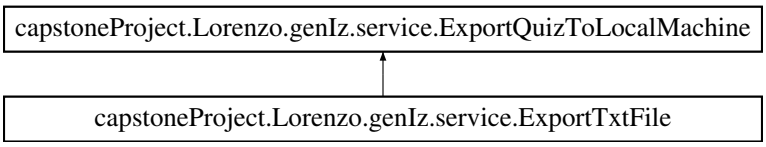
Implemented in capstoneProject.Lorenzo.genIz.service.ExportTxtFile.

The documentation for this interface was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/ExportQuizToLocalMachine.java

8.13 capstoneProject.Lorenzo.genIz.service.ExportTxtFile Class Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.ExportTxtFile:



Public Member Functions

- List< String > formatFileContent (List< ResponseDiscussionListDto > quizList)
- void saveFile (List< String > fileContent)

8.13.1 Detailed Description

Definition at line 21 of file ExportTxtFile.java.

8.13.2 Member Function Documentation

8.13.2.1 formatFileContent()

```
List< String > capstoneProject.Lorenzo.genIz.service.ExportTxtFile.formatFileContent (
            List< ResponseDiscussionListDto > quizList)
```

Implements capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine.

Definition at line 24 of file ExportTxtFile.java.

8.13.2.2 saveFile()

```
void capstoneProject.Lorenzo.genIz.service.ExportTxtFile.saveFile (
            List< String > fileContent)
```

Implements capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine.

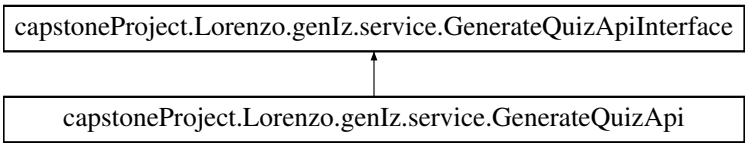
Definition at line 68 of file ExportTxtFile.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/ExportTxtFile.java

8.14 capstoneProject.Lorenzo.genIz.service.GenerateQuizApi Class Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.GenerateQuizApi:



Public Member Functions

- GenerateQuizApi (QuizDaoService quizDaoService)
- String createApiBody (String systemPrompt, String userText, Boolean useLocalModel)
- HttpRequest apiCallRequest (String requestBody, Boolean useLocalModel)
- ResponseDataDto getApiResponse (HttpRequest apiRequest)

Package Attributes

- QuizDaoService quizDaoService
- HttpClient httpClient = HttpClient.newHttpClient()
- Gson gson = new Gson()

Private Member Functions

- void retrieveOnlineModelSettings ()

Private Attributes

- String modelName
- String modelUrl
- ResponseOnlineModelDto responseOnlineModelDto

8.14.1 Detailed Description

Definition at line 23 of file GenerateQuizApi.java.

8.14.2 Constructor & Destructor Documentation

8.14.2.1 GenerateQuizApi()

```
capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.GenerateQuizApi (
            QuizDaoService quizDaoService)
```

Definition at line 28 of file GenerateQuizApi.java.

8.14.3 Member Function Documentation

8.14.3.1 apiCallRequest()

```
HttpRequest capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.apiCallRequest (
            String requestBody,
            Boolean useLocalModel)
```

Implements capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface.

Definition at line 83 of file GenerateQuizApi.java.

8.14.3.2 createApiBody()

```
String capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.createApiBody (
            String systemPrompt,
            String userText,
            Boolean useLocalModel)
```

Implements capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface.

Definition at line 63 of file GenerateQuizApi.java.

8.14.3.3 getApiResponse()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.getApiResponse (
            HttpRequest apiRequest)
```

Implements capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface.

Definition at line 125 of file GenerateQuizApi.java.

8.14.3.4 retrieveOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.retrieveOnlineModelSettings ()
[private]
```

Definition at line 43 of file GenerateQuizApi.java.

8.14.4 Member Data Documentation

8.14.4.1 gson

```
Gson capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.gson = new Gson()  [package]
```

Definition at line 58 of file GenerateQuizApi.java.

8.14.4.2 httpClient

```
HttpClient capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.httpClient = HttpClient.new↩
HttpClient()  [package]
```

Definition at line 55 of file GenerateQuizApi.java.

8.14.4.3 modelName

```
String capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.modelName  [private]
```

Definition at line 34 of file GenerateQuizApi.java.

8.14.4.4 modelUrl

```
String capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.modelUrl  [private]
```

Definition at line 37 of file GenerateQuizApi.java.

8.14.4.5 quizDaoService

```
QuizDaoService capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.quizDaoService  [package]
```

Definition at line 26 of file GenerateQuizApi.java.

8.14.4.6 responseOnlineModelDto

```
ResponseOnlineModelDto capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.responseOnline↩
ModelDto  [private]
```

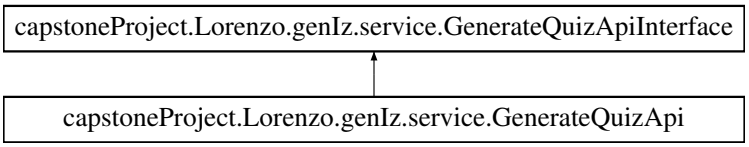
Definition at line 39 of file GenerateQuizApi.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/GenerateQuizApi.java

8.15 capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface Interface Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface:



Public Member Functions

- String createApiBody (String systemPrompt, String userText, Boolean useLocalModel)
- HttpRequest apiCallRequest (String requestBody, Boolean useLocalModel)
- ResponseDataDto getApiResponse (HttpRequest apiRequest)

8.15.1 Detailed Description

Definition at line 7 of file GenerateQuizApiInterface.java.

8.15.2 Member Function Documentation

8.15.2.1 apiCallRequest()

```
HttpRequest capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface.apiCallRequest (
            String requestBody,
            Boolean useLocalModel)
```

Implemented in capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.

8.15.2.2 createApiBody()

```
String capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface.createApiBody (
            String systemPrompt,
            String userText,
            Boolean useLocalModel)
```

Implemented in capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.

8.15.2.3 getApiResponse()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface.getApiResponse
(
            HttpRequest apiRequest)
```

Implemented in capstoneProject.Lorenzo.genIz.service.GenerateQuizApi.

The documentation for this interface was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/GenerateQuizApiInterface.java

8.16 capstoneProject.Lorenzo.genIz.controller.GenerateQuizController Class Reference

Public Member Functions

- GenerateQuizController (LocalModelApiFlowAggregator localModelApiFlowAggregator)
- ResponseDataDto generateQuizzesWithLocalModel (@RequestBody PostReqUserTextDto userTextDto)

Package Attributes

- LocalModelApiFlowAggregator localModelApiFlowAggregator

8.16.1 Detailed Description

Definition at line 15 of file GenerateQuizController.java.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 GenerateQuizController()

```
capstoneProject.Lorenzo.genIz.controller.GenerateQuizController.GenerateQuizController (
            LocalModelApiFlowAggregator localModelApiFlowAggregator)
```

Definition at line 20 of file GenerateQuizController.java.

8.16.3 Member Function Documentation

8.16.3.1 generateQuizzesWithLocalModel()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.controller.GenerateQuizController.generate↩
QuizzesWithLocalModel (
            @RequestBody PostReqUserTextDto userTextDto)
```

Definition at line 25 of file GenerateQuizController.java.

8.16.4 Member Data Documentation

8.16.4.1 localModelApiFlowAggregator

```
LocalModelApiFlowAggregator capstoneProject.Lorenzo.genIz.controller.GenerateQuizController.↩
localModelApiFlowAggregator  [package]
```

Definition at line 18 of file GenerateQuizController.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/controller/GenerateQuizController.java

8.17 capstoneProject.Lorenzo.genIz.GenIzApplication Class Reference

Static Public Member Functions

- static void main (String[ ] args)

8.17.1 Detailed Description

Definition at line 7 of file GenIzApplication.java.

8.17.2 Member Function Documentation

8.17.2.1 main()

```
void capstoneProject.Lorenzo.genIz.GenIzApplication.main (
            String[] args)  [static]
```

Definition at line 9 of file GenIzApplication.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/GenIzApplication.java

8.18 capstoneProject.Lorenzo.genIz.GenIzApplicationTests Class Reference

Package Functions

- void contextLoads ()

8.18.1 Detailed Description

Definition at line 7 of file GenIzApplicationTests.java.

8.18.2 Member Function Documentation

8.18.2.1 contextLoads()

```
void capstoneProject.Lorenzo.genIz.GenIzApplicationTests.contextLoads ()  [package]
```

Definition at line 10 of file GenIzApplicationTests.java.

The documentation for this class was generated from the following file:

- src/test/java/capstoneProject/Lorenzo/genIz/GenIzApplicationTests.java

8.19 capstoneProject.Lorenzo.genIz.global_exception_handler.GlobalExceptionHandler Class Reference

Public Member Functions

- ResponseEntity< ErrorResponseDto > handleIllegalArgumentException (IllegalArgumentException e)
- ResponseEntity< ErrorResponseDto > handleRuntimeException (RuntimeException e)
- ResponseEntity< ErrorResponseDto > handleNoResultException (NoResultException e)
- ResponseEntity< ErrorResponseDto > handleUnexpected (MethodArgumentTypeMismatchException ex)

8.19.1 Detailed Description

Definition at line 14 of file GlobalExceptionHandler.java.

8.19.2 Member Function Documentation

8.19.2.1 handleIllegalArgumentException()

```
ResponseEntity< ErrorResponseDto > capstoneProject.Lorenzo.genIz.global_exception_handler.↩
GlobalExceptionHandler.handleIllegalArgumentException (
            IllegalArgumentException e)
```

Definition at line 17 of file GlobalExceptionHandler.java.

8.19.2.2 handleNoResultException()

```
ResponseEntity< ErrorResponseDto > capstoneProject.Lorenzo.genIz.global_exception_handler.↩
GlobalExceptionHandler.handleNoResultException (
            NoResultException e)
```

Definition at line 29 of file GlobalExceptionHandler.java.

8.19.2.3 handleRuntimeException()

```
ResponseEntity< ErrorResponseDto > capstoneProject.Lorenzo.genIz.global_exception_handler.↩
GlobalExceptionHandler.handleRuntimeException (
            RuntimeException e)
```

Definition at line 23 of file GlobalExceptionHandler.java.

8.19.2.4 handleUnexpected()

```
ResponseEntity< ErrorResponseDto > capstoneProject.Lorenzo.genIz.global_exception_handler.↩
GlobalExceptionHandler.handleUnexpected (
            MethodArgumentTypeMismatchException ex)
```

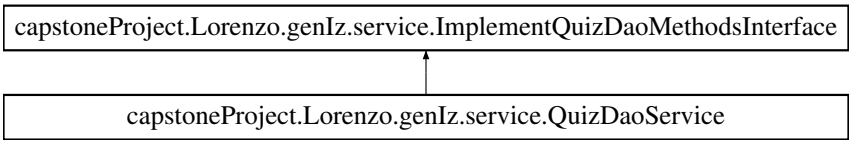
Definition at line 35 of file GlobalExceptionHandler.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/global_exception_handler/GlobalExceptionHandler.java

8.20 capstoneProject.Lorenzo.genIz.service.ImplementQuizDao↩ MethodsInterface Interface Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface:



Public Member Functions

- ResponseChatListDto saveChat (PostReqChatDto postReqChatDto)
- List< ResponseChatListDto > retrieveChats ()
- ResponseDiscussionListDto saveDiscussion (PostReqDiscussionDto postReqDiscussionDto)
- List< ResponseDiscussionListDto > retrieveDiscussions (Integer chatId)
- ResponseModelSettingDto manageModelSettings (PostReqModelSetting postReqModelSetting)
- ResponseModelSettingDto retrieveModelSettings ()
- void deleteChat (Integer chatId)
- ResponseChatListDto updateChat (PostReqChatDto postReqChatDto)
- void deleteDiscussion (Integer discussionId)
- void saveNewOnlineModelSettings (PostReqOnlineModelDto postReqOnlineModelDto)
- ResponseOnlineModelDto retrieveOnlineModelSettings ()
- void deleteOnlineModelSettings ()

8.20.1 Detailed Description

Definition at line 15 of file ImplementQuizDaoMethodsInterface.java.

8.20.2 Member Function Documentation

8.20.2.1 deleteChat()

```
void capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.deleteChat (
            Integer chatId)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.2 deleteDiscussion()

```
void capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.deleteDiscussion (
            Integer discussionId)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.3 deleteOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.deleteOnline↩
ModelSettings ()
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.4 manageModelSettings()

```
ResponseModelSettingDto capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.↩
manageModelSettings (
            PostReqModelSetting postReqModelSetting)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.5 retrieveChats()

```
List< ResponseChatListDto > capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethods↩
Interface.retrieveChats ()
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.6 retrieveDiscussions()

```
List< ResponseDiscussionListDto > capstoneProject.Lorenzo.genIz.service.ImplementQuizDao↩
MethodsInterface.retrieveDiscussions (
            Integer chatId)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.7 retrieveModelSettings()

```
ResponseModelSettingDto capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.↩
retrieveModelSettings ()
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.8 retrieveOnlineModelSettings()

```
ResponseOnlineModelDto capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.↩
retrieveOnlineModelSettings ()
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.9 saveChat()

```
ResponseChatListDto capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.↩
saveChat (
            PostReqChatDto postReqChatDto)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.10 saveDiscussion()

```
ResponseDiscussionListDto capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethods↩
Interface.saveDiscussion (
            PostReqDiscussionDto postReqDiscussionDto)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.11 saveNewOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.saveNewOnline↩
ModelSettings (
            PostReqOnlineModelDto postReqOnlineModelDto)
```

Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

8.20.2.12 updateChat()

```
ResponseChatListDto capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.↩
updateChat (
            PostReqChatDto postReqChatDto)
```

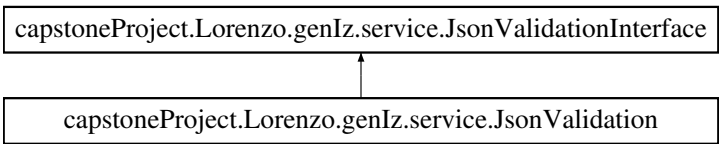
Implemented in capstoneProject.Lorenzo.genIz.service.QuizDaoService.

The documentation for this interface was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/ImplementQuizDaoMethodsInterface.java

8.21 capstoneProject.Lorenzo.genIz.service.JsonValidation Class Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.JsonValidation:



Public Member Functions

- JsonObject convertStringToJsonObject (ResponseDataDto quizDataDto)
- ResponseDataDto validateJsonObject (JsonObject objectToValidate, ResponseDataDto quizDataDto)

8.21.1 Detailed Description

Definition at line 13 of file JsonValidation.java.

8.21.2 Member Function Documentation

8.21.2.1 convertStringToJsonObject()

```
JsonObject capstoneProject.Lorenzo.genIz.service.JsonValidation.convertStringToJsonObject (
            ResponseDataDto quizDataDto)
```

Implements capstoneProject.Lorenzo.genIz.service.JsonValidationInterface.

Definition at line 17 of file JsonValidation.java.

8.21.2.2 validateJsonObject()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.JsonValidation.validateJsonObject (
            JsonObject objectToValidate,
            ResponseDataDto quizDataDto)
```

Implements capstoneProject.Lorenzo.genIz.service.JsonValidationInterface.

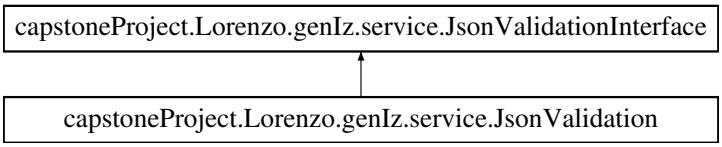
Definition at line 31 of file JsonValidation.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/JsonValidation.java

8.22 capstoneProject.Lorenzo.genIz.service.JsonValidationInterface Interface Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.JsonValidationInterface:



Public Member Functions

- JsonObject convertStringToJsonObject (ResponseDataDto quizDataDto)
- ResponseDataDto validateJsonObject (JsonObject objectToValidate, ResponseDataDto quizDataDto)

8.22.1 Detailed Description

Definition at line 7 of file JsonValidationInterface.java.

8.22.2 Member Function Documentation

8.22.2.1 convertStringToJsonObject()

```
JsonObject capstoneProject.Lorenzo.genIz.service.JsonValidationInterface.convertStringToJson↩
Object (
            ResponseDataDto quizDataDto)
```

Implemented in capstoneProject.Lorenzo.genIz.service.JsonValidation.

8.22.2.2 validateJsonObject()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.JsonValidationInterface.validateJson↩
Object (
            JsonObject objectToValidate,
            ResponseDataDto quizDataDto)
```

Implemented in capstoneProject.Lorenzo.genIz.service.JsonValidation.

The documentation for this interface was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/JsonValidationInterface.java

8.23 capstoneProject.Lorenzo.genIz.JsonValidationTest Class Reference

Package Functions

- void testCorrectExecutionOfTheValidationSteps ()
- void testValidationStepsWithMissingQuizzesArray ()
- void testValidationStepsWithMissingOrMispelledQuestionSection ()
- void testValidationStepsWithMissingOrMispelledOptionsSection ()
- void testValidationStepsWithMissingOrMispelledOptionLetter ()
- void testValidationStepsWithMissingOrMispelledExplanationSection ()
- void testValidationStepsWithMissingOrMispelledAnswerSection ()

8.23.1 Detailed Description

Definition at line 10 of file JsonValidationTest.java.

8.23.2 Member Function Documentation

8.23.2.1 testCorrectExecutionOfTheValidationSteps()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testCorrectExecutionOfTheValidationSteps
() [package]
```

Definition at line 14 of file JsonValidationTest.java.

8.23.2.2 testValidationStepsWithMissingOrMispelledAnswerSection()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testValidationStepsWithMissingOrMispelled↩
AnswerSection () [package]
```

Definition at line 103 of file JsonValidationTest.java.

8.23.2.3 testValidationStepsWithMissingOrMispelledExplanationSection()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testValidationStepsWithMissingOrMispelled↩
ExplanationSection () [package]
```

Definition at line 88 of file JsonValidationTest.java.

8.23.2.4 testValidationStepsWithMissingOrMispelledOptionLetter()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testValidationStepsWithMissingOrMispelled↩
OptionLetter () [package]
```

Definition at line 73 of file JsonValidationTest.java.

8.23.2.5 testValidationStepsWithMissingOrMispelledOptionsSection()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testValidationStepsWithMissingOrMispelled↩
OptionsSection () [package]
```

Definition at line 58 of file JsonValidationTest.java.

8.23.2.6 testValidationStepsWithMissingOrMispelledQuestionSection()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testValidationStepsWithMissingOrMispelled↩
QuestionSection () [package]
```

Definition at line 43 of file JsonValidationTest.java.

8.23.2.7 testValidationStepsWithMissingQuizzesArray()

```
void capstoneProject.Lorenzo.genIz.JsonValidationTest.testValidationStepsWithMissingQuizzes↩
Array () [package]
```

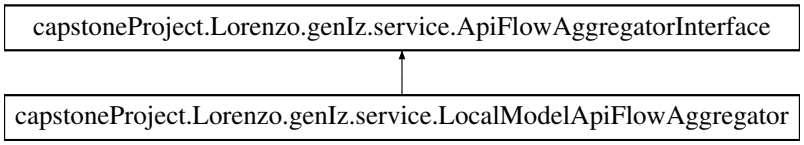
Definition at line 28 of file JsonValidationTest.java.

The documentation for this class was generated from the following file:

- src/test/java/capstoneProject/Lorenzo/genIz/JsonValidationTest.java

8.24 capstoneProject.Lorenzo.genIz.service.LocalModelApiFlow↩ Aggregator Class Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator:



Public Member Functions

- LocalModelApiFlowAggregator (GenerateQuizApi generateQuizApi, JsonValidation jsonValidation)
- ResponseDataDto performApiCall (PostReqUserTextDto postReqUserTextDto)
- ResponseDataDto validateApiCallResult (ResponseDataDto tempDTO)

Package Attributes

- GenerateQuizApi generateQuizApi
- JsonValidation jsonValidation

8.24.1 Detailed Description

Definition at line 14 of file LocalModelApiFlowAggregator.java.

8.24.2 Constructor & Destructor Documentation

8.24.2.1 LocalModelApiFlowAggregator()

```
capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.LocalModelApiFlowAggregator
(
            GenerateQuizApi generateQuizApi,
            JsonValidation jsonValidation)
```

Definition at line 20 of file LocalModelApiFlowAggregator.java.

8.24.3 Member Function Documentation

8.24.3.1 performApiCall()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.performApi↩
Call (
            PostReqUserTextDto postReqUserTextDto)
```

Implements capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface.

Definition at line 26 of file LocalModelApiFlowAggregator.java.

8.24.3.2 validateApiCallResult()

```
ResponseDataDto capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.validate↩
ApiCallResult (
            ResponseDataDto tempDTO)
```

Implements capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface.

Definition at line 41 of file LocalModelApiFlowAggregator.java.

8.24.4 Member Data Documentation

8.24.4.1 generateQuizApi

```
GenerateQuizApi capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.generate↩
QuizApi  [package]
```

Definition at line 17 of file LocalModelApiFlowAggregator.java.

8.24.4.2 jsonValidation

```
JsonValidation capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator.jsonValidation
[package]
```

Definition at line 18 of file LocalModelApiFlowAggregator.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/LocalModelApiFlowAggregator.java

8.25 capstoneProject.Lorenzo.genIz.api_format.response.map_model_↩ response.Message Class Reference

Public Member Functions

- String getContent ()
- void setContent (String content)

Private Attributes

- String content

8.25.1 Detailed Description

Definition at line 3 of file Message.java.

8.25.2 Member Function Documentation

8.25.2.1 getContent()

```
String capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Message.getContent
()
```

Definition at line 6 of file Message.java.

8.25.2.2 setContent()

```
void capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Message.setContent (
            String content)
```

Definition at line 10 of file Message.java.

8.25.3 Member Data Documentation

8.25.3.1 content

```
String capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Message.content
[private]
```

Definition at line 4 of file Message.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/map_model_response/Message.java

8.26 capstoneProject.Lorenzo.genIz.controller.OnlineModelController Class Reference

Public Member Functions

- OnlineModelController (ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface)
- void saveNewSettings (@RequestBody PostReqOnlineModelDto postReqOnlineModelDto)
- void deleteSettings ()

Package Attributes

- ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface

8.26.1 Detailed Description

Definition at line 14 of file OnlineModelController.java.

8.26.2 Constructor & Destructor Documentation

8.26.2.1 OnlineModelController()

```
capstoneProject.Lorenzo.genIz.controller.OnlineModelController.OnlineModelController (
            ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface)
```

Definition at line 19 of file OnlineModelController.java.

8.26.3 Member Function Documentation

8.26.3.1 deleteSettings()

```
void capstoneProject.Lorenzo.genIz.controller.OnlineModelController.deleteSettings ()
```

Definition at line 31 of file OnlineModelController.java.

8.26.3.2 saveNewSettings()

```
void capstoneProject.Lorenzo.genIz.controller.OnlineModelController.saveNewSettings (
            @RequestBody PostReqOnlineModelDto postReqOnlineModelDto)
```

Definition at line 25 of file OnlineModelController.java.

8.26.4 Member Data Documentation

8.26.4.1 implementQuizDaoMethodsInterface

```
ImplementQuizDaoMethodsInterface capstoneProject.Lorenzo.genIz.controller.OnlineModelController.↩
implementQuizDaoMethodsInterface  [package]
```

Definition at line 17 of file OnlineModelController.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/controller/OnlineModelController.java

8.27 capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity Class Reference

Public Member Functions

- Integer getId ()
- void setId (Integer id)
- String getModelUrl ()
- void setModelUrl (String modelUrl)
- String getModelName ()
- void setModelName (String modelName)
- String getApiKey ()
- void setApiKey (String apiKey)

Private Attributes

- Integer id
- String modelUrl
- String modelName
- String apiKey

8.27.1 Detailed Description

Definition at line 12 of file OnlineModelEntity.java.

8.27.2 Member Function Documentation

8.27.2.1 getApiKey()

```
String capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.getApiKey ()
```

Definition at line 52 of file OnlineModelEntity.java.

8.27.2.2 getId()

```
Integer capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.getId ()
```

Definition at line 28 of file OnlineModelEntity.java.

8.27.2.3 getModelName()

```
String capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.getModelName ()
```

Definition at line 44 of file OnlineModelEntity.java.

8.27.2.4 getModelUrl()

```
String capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.getModelUrl ()
```

Definition at line 36 of file OnlineModelEntity.java.

8.27.2.5 setApiKey()

```
void capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.setApiKey (
            String apiKey)
```

Definition at line 56 of file OnlineModelEntity.java.

8.27.2.6 setId()

```
void capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.setId (
            Integer id)
```

Definition at line 32 of file OnlineModelEntity.java.

8.27.2.7 setModelName()

```
void capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.setModelName (
            String modelName)
```

Definition at line 48 of file OnlineModelEntity.java.

8.27.2.8 setModelUrl()

```
void capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.setModelUrl (
            String modelUrl)
```

Definition at line 40 of file OnlineModelEntity.java.

8.27.3 Member Data Documentation

8.27.3.1 apiKey

```
String capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.apiKey  [private]
```

Definition at line 26 of file OnlineModelEntity.java.

8.27.3.2 id

```
Integer capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.id  [private]
```

Definition at line 17 of file OnlineModelEntity.java.

8.27.3.3 modelName

```
String capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.modelName  [private]
```

Definition at line 23 of file OnlineModelEntity.java.

8.27.3.4 modelUrl

```
String capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity.modelUrl  [private]
```

Definition at line 20 of file OnlineModelEntity.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/entity/OnlineModelEntity.java

8.28 capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto Class Reference

Public Member Functions

- String getChatName ()
- void setChatName (String chatName)
- Integer getChat_id ()
- void setChat_id (Integer chat_id)

Private Attributes

- String chatName
- Integer chat_id

8.28.1 Detailed Description

Definition at line 3 of file PostReqChatDto.java.

8.28.2 Member Function Documentation

8.28.2.1 getChat_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto.getChat_id ()
```

Definition at line 15 of file PostReqChatDto.java.

8.28.2.2 getChatName()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto.getChatName ()
```

Definition at line 7 of file PostReqChatDto.java.

8.28.2.3 setChat_id()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto.setChat_id (
            Integer chat_id)
```

Definition at line 19 of file PostReqChatDto.java.

8.28.2.4 setChatName()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto.setChatName (
            String chatName)
```

Definition at line 11 of file PostReqChatDto.java.

8.28.3 Member Data Documentation

8.28.3.1 chat_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto.chat_id  [private]
```

Definition at line 5 of file PostReqChatDto.java.

8.28.3.2 chatName

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto.chatName  [private]
```

Definition at line 4 of file PostReqChatDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/PostReqChatDto.java

8.29 capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReq↩ DiscussionDto Class Reference

Public Member Functions

- Integer getChat_id ()
- void setChat_id (Integer chat_id)
- String getUser_pdf_name ()
- void setUser_pdf_name (String user_pdf_name)
- String getQuiz_content ()
- void setQuiz_content (String quiz_content)

Private Attributes

- Integer chat_id
- String user_pdf_name
- String quiz_content

8.29.1 Detailed Description

Definition at line 3 of file PostReqDiscussionDto.java.

8.29.2 Member Function Documentation

8.29.2.1 getChat_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.getChat_id ()
```

Definition at line 8 of file PostReqDiscussionDto.java.

8.29.2.2 getQuiz_content()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.getQuiz_content ()
```

Definition at line 20 of file PostReqDiscussionDto.java.

8.29.2.3 getUser_pdf_name()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.getUser_pdf_name ()
```

Definition at line 14 of file PostReqDiscussionDto.java.

8.29.2.4 setChat_id()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.setChat_id (
            Integer chat_id)
```

Definition at line 11 of file PostReqDiscussionDto.java.

8.29.2.5 setQuiz_content()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.setQuiz_content (
            String quiz_content)
```

Definition at line 23 of file PostReqDiscussionDto.java.

8.29.2.6 setUser_pdf_name()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.setUser_pdf_name (
            String user_pdf_name)
```

Definition at line 17 of file PostReqDiscussionDto.java.

8.29.3 Member Data Documentation

8.29.3.1 chat_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.chat_id  [private]
```

Definition at line 4 of file PostReqDiscussionDto.java.

8.29.3.2 quiz_content

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.quiz_content  [private]
```

Definition at line 6 of file PostReqDiscussionDto.java.

8.29.3.3 user_pdf_name

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto.user_pdf_name  [private]
```

Definition at line 5 of file PostReqDiscussionDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/PostReqDiscussionDto.java

8.30 capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModel↩ Setting Class Reference

Public Member Functions

- Boolean getModelSetting ()
- void setModelSetting (Boolean modelSetting)

Private Attributes

- Boolean modelSetting

8.30.1 Detailed Description

Definition at line 3 of file PostReqModelSetting.java.

8.30.2 Member Function Documentation

8.30.2.1 getModelSetting()

```
Boolean capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting.getModelSetting ()
```

Definition at line 6 of file PostReqModelSetting.java.

8.30.2.2 setModelSetting()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting.setModelSetting (
            Boolean modelSetting)
```

Definition at line 10 of file PostReqModelSetting.java.

8.30.3 Member Data Documentation

8.30.3.1 modelSetting

```
Boolean capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting.modelSetting  [private]
```

Definition at line 4 of file PostReqModelSetting.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/PostReqModelSetting.java

8.31 capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnline↩ ModelDto Class Reference

Public Member Functions

- String getModel_url ()
- void setModel_url (String model_url)
- String getModel_name ()
- void setModel_name (String model_name)
- String getApi_key ()
- void setApi_key (String api_key)

Private Attributes

- String model_url
- String model_name
- String api_key

8.31.1 Detailed Description

Definition at line 3 of file PostReqOnlineModelDto.java.

8.31.2 Member Function Documentation

8.31.2.1 getApi_key()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.getApi_key ()
```

Definition at line 20 of file PostReqOnlineModelDto.java.

8.31.2.2 getModel_name()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.getModel_name ()
```

Definition at line 14 of file PostReqOnlineModelDto.java.

8.31.2.3 getModel_url()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.getModel_url ()
```

Definition at line 8 of file PostReqOnlineModelDto.java.

8.31.2.4 setApi_key()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.setApi_key (
            String api_key)
```

Definition at line 23 of file PostReqOnlineModelDto.java.

8.31.2.5 setModel_name()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.setModel_name (
            String model_name)
```

Definition at line 17 of file PostReqOnlineModelDto.java.

8.31.2.6 setModel_url()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.setModel_url (
            String model_url)
```

Definition at line 11 of file PostReqOnlineModelDto.java.

8.31.3 Member Data Documentation

8.31.3.1 api_key

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.api_key  [private]
```

Definition at line 6 of file PostReqOnlineModelDto.java.

8.31.3.2 model_name

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.model_name  [private]
```

Definition at line 5 of file PostReqOnlineModelDto.java.

8.31.3.3 model_url

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto.model_url  [private]
```

Definition at line 4 of file PostReqOnlineModelDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/PostReqOnlineModelDto.java

8.32 capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUser↩ TextDto Class Reference

Public Member Functions

- String getUserText ()
- void setUserText (String userText)
- Boolean getUseLocalModel ()
- void setUseLocalModel (Boolean useLocalModel)

Private Attributes

- String userText
- Boolean useLocalModel

8.32.1 Detailed Description

Definition at line 3 of file PostReqUserTextDto.java.

8.32.2 Member Function Documentation

8.32.2.1 getUseLocalModel()

```
Boolean capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto.getUseLocalModel ()
```

Definition at line 15 of file PostReqUserTextDto.java.

8.32.2.2 getUserText()

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto.getUserText ()
```

Definition at line 7 of file PostReqUserTextDto.java.

8.32.2.3 setUseLocalModel()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto.setUseLocalModel (
            Boolean useLocalModel)
```

Definition at line 19 of file PostReqUserTextDto.java.

8.32.2.4 setUserText()

```
void capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto.setUserText (
            String userText)
```

Definition at line 11 of file PostReqUserTextDto.java.

8.32.3 Member Data Documentation

8.32.3.1 useLocalModel

```
Boolean capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto.useLocalModel  [private]
```

Definition at line 5 of file PostReqUserTextDto.java.

8.32.3.2 userText

```
String capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto.userText  [private]
```

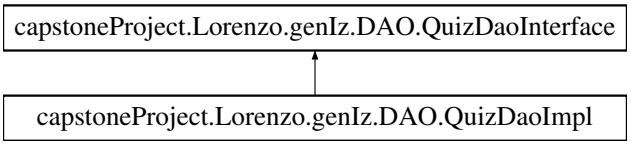
Definition at line 4 of file PostReqUserTextDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/PostReqUserTextDto.java

8.33 capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl Class Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl:



Public Member Functions

- QuizDaoImpl (EntityManager entityManager)
- UserDataDTO manageUser (OAuth2AuthenticationToken token)
- ChatDataDto saveChat (UserDataDTO userDataDTO, PostReqChatDto postReqChatDto)
- List< ChatDataDto > retrieveChats (UserDataDTO userDataDTO)
- DiscussionDataDto saveDiscussion (PostReqDiscussionDto postReqDiscussionDto)
- List< DiscussionDataDto > retrieveDiscussions (Integer chatId)
- UserDataDTO manageModelSettings (UserDataDTO userDataDTO, PostReqModelSetting postReqModel↩ Setting)
- UserDataDTO retrieveModelSettings (UserDataDTO userDataDTO)
- void deleteChat (Integer chatId)
- ChatDataDto updateChat (PostReqChatDto postReqChatDto)
- void deleteDiscussion (Integer discussionId)
- void saveNewOnlineModelSettings (PostReqOnlineModelDto postReqOnlineModelDto)
- ResponseOnlineModelDto retrieveOnlineModelSettings ()
- void deleteOnlineModelSettings ()

Package Attributes

- EntityManager entityManager

8.33.1 Detailed Description

Definition at line 29 of file QuizDaoImpl.java.

8.33.2 Constructor & Destructor Documentation

8.33.2.1 QuizDaoImpl()

```
capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.QuizDaoImpl (
            EntityManager entityManager)
```

Definition at line 34 of file QuizDaoImpl.java.

8.33.3 Member Function Documentation

8.33.3.1 deleteChat()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.deleteChat (
            Integer chatId)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 282 of file QuizDaoImpl.java.

8.33.3.2 deleteDiscussion()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.deleteDiscussion (
            Integer discussionId)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 335 of file QuizDaoImpl.java.

8.33.3.3 deleteOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.deleteOnlineModelSettings ()
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 412 of file QuizDaoImpl.java.

8.33.3.4 manageModelSettings()

```
UserDataDTO capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.manageModelSettings (
            UserDataDTO userDataDTO,
            PostReqModelSetting postReqModelSetting)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 242 of file QuizDaoImpl.java.

8.33.3.5 manageUser()

```
UserDataDTO capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.manageUser (
            OAuth2AuthenticationToken token)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 42 of file QuizDaoImpl.java.

8.33.3.6 retrieveChats()

```
List< ChatDataDto > capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.retrieveChats (
            UserDataDTO userDataDTO)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 115 of file QuizDaoImpl.java.

8.33.3.7 retrieveDiscussions()

```
List< DiscussionDataDto > capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.retrieveDiscussions (
            Integer chatId)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 195 of file QuizDaoImpl.java.

8.33.3.8 retrieveModelSettings()

```
UserDataDTO capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.retrieveModelSettings (
            UserDataDTO userDataDTO)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 265 of file QuizDaoImpl.java.

8.33.3.9 retrieveOnlineModelSettings()

```
ResponseOnlineModelDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.retrieveOnlineModel↩
Settings ()
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 388 of file QuizDaoImpl.java.

8.33.3.10 saveChat()

```
ChatDataDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.saveChat (
            UserDataDTO userDataDTO,
            PostReqChatDto postReqChatDto)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 83 of file QuizDaoImpl.java.

8.33.3.11 saveDiscussion()

```
DiscussionDataDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.saveDiscussion (
            PostReqDiscussionDto postReqDiscussionDto)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 150 of file QuizDaoImpl.java.

8.33.3.12 saveNewOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.saveNewOnlineModelSettings (
            PostReqOnlineModelDto postReqOnlineModelDto)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 357 of file QuizDaoImpl.java.

8.33.3.13 updateChat()

```
ChatDataDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.updateChat (
            PostReqChatDto postReqChatDto)
```

Implements capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.

Definition at line 305 of file QuizDaoImpl.java.

8.33.4 Member Data Documentation

8.33.4.1 entityManager

```
EntityManager capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.entityManager  [package]
```

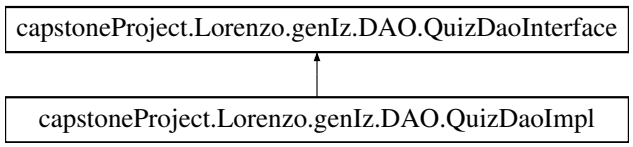
Definition at line 32 of file QuizDaoImpl.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DAO/QuizDaoImpl.java

8.34 capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface Interface Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface:



Public Member Functions

- UserDataDTO manageUser (OAuth2AuthenticationToken token)
- ChatDataDto saveChat (UserDataDTO userDataDTO, PostReqChatDto postReqChatDto)
- List< ChatDataDto > retrieveChats (UserDataDTO userDataDTO)
- DiscussionDataDto saveDiscussion (PostReqDiscussionDto postReqDiscussionDto)
- List< DiscussionDataDto > retrieveDiscussions (Integer chatId)
- UserDataDTO manageModelSettings (UserDataDTO userDataDTO, PostReqModelSetting postReqModel↩ Setting)
- UserDataDTO retrieveModelSettings (UserDataDTO userDataDTO)
- void deleteChat (Integer chatId)
- ChatDataDto updateChat (PostReqChatDto postReqChatDto)
- void deleteDiscussion (Integer discussionId)
- void saveNewOnlineModelSettings (PostReqOnlineModelDto postReqOnlineModelDto)
- ResponseOnlineModelDto retrieveOnlineModelSettings ()
- void deleteOnlineModelSettings ()

8.34.1 Detailed Description

Definition at line 16 of file QuizDaoInterface.java.

8.34.2 Member Function Documentation

8.34.2.1 deleteChat()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.deleteChat (
            Integer chatId)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.2 deleteDiscussion()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.deleteDiscussion (
            Integer discussionId)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.3 deleteOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.deleteOnlineModelSettings ()
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.4 manageModelSettings()

```
UserDataDTO capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.manageModelSettings (
            UserDataDTO userDataDTO,
            PostReqModelSetting postReqModelSetting)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.5 manageUser()

```
UserDataDTO capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.manageUser (
            OAuth2AuthenticationToken token)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.6 retrieveChats()

```
List< ChatDataDto > capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.retrieveChats (
            UserDataDTO userDataDTO)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.7 retrieveDiscussions()

```
List< DiscussionDataDto > capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.retrieveDiscussions
(
            Integer chatId)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.8 retrieveModelSettings()

```
UserDataDTO capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.retrieveModelSettings (
            UserDataDTO userDataDTO)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.9 retrieveOnlineModelSettings()

```
ResponseOnlineModelDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.retrieveOnline↩
ModelSettings ()
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.10 saveChat()

```
ChatDataDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.saveChat (
            UserDataDTO userDataDTO,
            PostReqChatDto postReqChatDto)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.11 saveDiscussion()

```
DiscussionDataDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.saveDiscussion (
            PostReqDiscussionDto postReqDiscussionDto)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.12 saveNewOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.saveNewOnlineModelSettings (
            PostReqOnlineModelDto postReqOnlineModelDto)
```

Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

8.34.2.13 updateChat()

```
ChatDataDto capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface.updateChat (
            PostReqChatDto postReqChatDto)
```

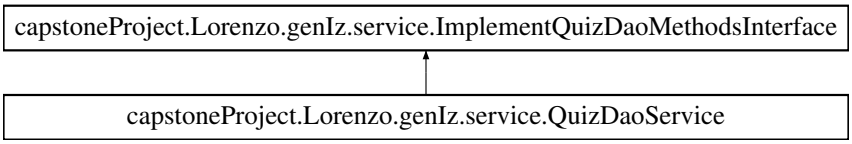
Implemented in capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl.

The documentation for this interface was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DAO/QuizDaoInterface.java

8.35 capstoneProject.Lorenzo.genIz.service.QuizDaoService Class Reference

Inheritance diagram for capstoneProject.Lorenzo.genIz.service.QuizDaoService:



Public Member Functions

- QuizDaoService (QuizDaoImpl quizDaoImpl)
- OAuth2AuthenticationToken retrieveToken ()
- ResponseChatListDto saveChat (PostReqChatDto postReqChatDto)
- List< ResponseChatListDto > retrieveChats ()
- ResponseDiscussionListDto saveDiscussion (PostReqDiscussionDto postReqDiscussionDto)
- List< ResponseDiscussionListDto > retrieveDiscussions (Integer chatId)
- ResponseModelSettingDto manageModelSettings (PostReqModelSetting postReqModelSetting)
- ResponseModelSettingDto retrieveModelSettings ()
- void deleteChat (Integer chatId)
- ResponseChatListDto updateChat (PostReqChatDto postReqChatDto)
- void deleteDiscussion (Integer discussionId)
- void saveNewOnlineModelSettings (PostReqOnlineModelDto postReqOnlineModelDto)
- ResponseOnlineModelDto retrieveOnlineModelSettings ()
- void deleteOnlineModelSettings ()

Package Attributes

- QuizDaoImpl quizDaoImpl

8.35.1 Detailed Description

Definition at line 24 of file QuizDaoService.java.

8.35.2 Constructor & Destructor Documentation

8.35.2.1 QuizDaoService()

```
capstoneProject.Lorenzo.genIz.service.QuizDaoService.QuizDaoService (
            QuizDaoImpl quizDaoImpl)
```

Definition at line 29 of file QuizDaoService.java.

8.35.3 Member Function Documentation

8.35.3.1 deleteChat()

```
void capstoneProject.Lorenzo.genIz.service.QuizDaoService.deleteChat (
            Integer chatId)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 162 of file QuizDaoService.java.

8.35.3.2 deleteDiscussion()

```
void capstoneProject.Lorenzo.genIz.service.QuizDaoService.deleteDiscussion (
            Integer discussionId)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 192 of file QuizDaoService.java.

8.35.3.3 deleteOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.service.QuizDaoService.deleteOnlineModelSettings ()
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 224 of file QuizDaoService.java.

8.35.3.4 manageModelSettings()

```
ResponseModelSettingDto capstoneProject.Lorenzo.genIz.service.QuizDaoService.manageModel↩
Settings (
            PostReqModelSetting postReqModelSetting)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 125 of file QuizDaoService.java.

8.35.3.5 retrieveChats()

```
List< ResponseChatListDto > capstoneProject.Lorenzo.genIz.service.QuizDaoService.retrieveChats
()
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 59 of file QuizDaoService.java.

8.35.3.6 retrieveDiscussions()

```
List< ResponseDiscussionListDto > capstoneProject.Lorenzo.genIz.service.QuizDaoService.↩
retrieveDiscussions (
            Integer chatId)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 102 of file QuizDaoService.java.

8.35.3.7 retrieveModelSettings()

```
ResponseModelSettingDto capstoneProject.Lorenzo.genIz.service.QuizDaoService.retrieveModel↩
Settings ()
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 143 of file QuizDaoService.java.

8.35.3.8 retrieveOnlineModelSettings()

```
ResponseOnlineModelDto capstoneProject.Lorenzo.genIz.service.QuizDaoService.retrieveOnline↩
ModelSettings ()
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 211 of file QuizDaoService.java.

8.35.3.9 retrieveToken()

```
OAuth2AuthenticationToken capstoneProject.Lorenzo.genIz.service.QuizDaoService.retrieveToken
()
```

Definition at line 34 of file QuizDaoService.java.

8.35.3.10 saveChat()

```
ResponseChatListDto capstoneProject.Lorenzo.genIz.service.QuizDaoService.saveChat (
            PostReqChatDto postReqChatDto)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 39 of file QuizDaoService.java.

8.35.3.11 saveDiscussion()

```
ResponseDiscussionListDto capstoneProject.Lorenzo.genIz.service.QuizDaoService.saveDiscussion
(
            PostReqDiscussionDto postReqDiscussionDto)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 83 of file QuizDaoService.java.

8.35.3.12 saveNewOnlineModelSettings()

```
void capstoneProject.Lorenzo.genIz.service.QuizDaoService.saveNewOnlineModelSettings (
            PostReqOnlineModelDto postReqOnlineModelDto)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 204 of file QuizDaoService.java.

8.35.3.13 updateChat()

```
ResponseChatListDto capstoneProject.Lorenzo.genIz.service.QuizDaoService.updateChat (
            PostReqChatDto postReqChatDto)
```

Implements capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface.

Definition at line 174 of file QuizDaoService.java.

8.35.4 Member Data Documentation

8.35.4.1 quizDaoImpl

```
QuizDaoImpl capstoneProject.Lorenzo.genIz.service.QuizDaoService.quizDaoImpl  [package]
```

Definition at line 27 of file QuizDaoService.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/service/QuizDaoService.java

8.36 capstoneProject.Lorenzo.genIz.api_format.request.request_↩ parameters.QuizGenParameter Class Reference

Static Public Member Functions

- static String getSystemprompt ()

Static Public Attributes

- static final String SYSTEMPROMPT

8.36.1 Detailed Description

Definition at line 3 of file QuizGenParameter.java.

8.36.2 Member Function Documentation

8.36.2.1 getSystemprompt()

```
String capstoneProject.Lorenzo.genIz.api_format.request.request_parameters.QuizGenParameter.↩
getSystemprompt () [static]
```

Definition at line 70 of file QuizGenParameter.java.

8.36.3 Member Data Documentation

8.36.3.1 SYSTEMPROMPT

```
final String capstoneProject.Lorenzo.genIz.api_format.request.request_parameters.QuizGen↩
Parameter.SYSTEMPROMPT [static]
```

Definition at line 5 of file QuizGenParameter.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/api_format/request/request_parameters/QuizGenParameter.java

8.37 capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChat↩ ListDto Class Reference

Public Member Functions

- Integer getChat_id ()
- void setChat_id (Integer chat_id)
- String getChat_name ()
- void setChat_name (String chat_name)

Private Attributes

- Integer chat_id
- String chat_name

8.37.1 Detailed Description

Definition at line 3 of file ResponseChatListDto.java.

8.37.2 Member Function Documentation

8.37.2.1 getChat_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto.getChat_id ()
```

Definition at line 7 of file ResponseChatListDto.java.

8.37.2.2 getChat_name()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto.getChat_name ()
```

Definition at line 13 of file ResponseChatListDto.java.

8.37.2.3 setChat_id()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto.setChat_id (
            Integer chat_id)
```

Definition at line 10 of file ResponseChatListDto.java.

8.37.2.4 setChat_name()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto.setChat_name (
            String chat_name)
```

Definition at line 16 of file ResponseChatListDto.java.

8.37.3 Member Data Documentation

8.37.3.1 chat_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto.chat_id  [private]
```

Definition at line 4 of file ResponseChatListDto.java.

8.37.3.2 chat_name

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto.chat_name  [private]
```

Definition at line 5 of file ResponseChatListDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/ResponseChatListDto.java

8.38 capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto Class Reference

Public Member Functions

- ResponseDataDto (String llmOutput, Integer tokenCount)
- String getLlmOutput ()
- void setLlmOutput (String llmOutput)
- Integer getTokenCount ()
- void setTokenCount (Integer tokenCount)

Private Attributes

- String llmOutput
- Integer tokenCount

8.38.1 Detailed Description

Definition at line 3 of file ResponseDataDto.java.

8.38.2 Constructor & Destructor Documentation

8.38.2.1 ResponseDataDto()

```
capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.ResponseDataDto (
            String llmOutput,
            Integer tokenCount)
```

Definition at line 7 of file ResponseDataDto.java.

8.38.3 Member Function Documentation

8.38.3.1 getLlmOutput()

```
String capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.getLlmOutput ()
```

Definition at line 12 of file ResponseDataDto.java.

8.38.3.2 getTokenCount()

```
Integer capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.getTokenCount ()
```

Definition at line 20 of file ResponseDataDto.java.

8.38.3.3 setLlmOutput()

```
void capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.setLlmOutput (
            String llmOutput)
```

Definition at line 16 of file ResponseDataDto.java.

8.38.3.4 setTokenCount()

```
void capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.setTokenCount (
            Integer tokenCount)
```

Definition at line 24 of file ResponseDataDto.java.

8.38.4 Member Data Documentation

8.38.4.1 llmOutput

```
String capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.llmOutput  [private]
```

Definition at line 4 of file ResponseDataDto.java.

8.38.4.2 tokenCount

```
Integer capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto.tokenCount  [private]
```

Definition at line 5 of file ResponseDataDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/ResponseDataDto.java

8.39 capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto Class Reference

Public Member Functions

- Integer getDiscussion_id ()
- void setDiscussion_id (Integer discussion_id)
- String getUser_pdf_name ()
- void setUser_pdf_name (String user_pdf_name)
- String getQuiz_content ()
- void setQuiz_content (String quiz_content)

Private Attributes

- Integer discussion_id
- String user_pdf_name
- String quiz_content

8.39.1 Detailed Description

Definition at line 3 of file ResponseDiscussionListDto.java.

8.39.2 Member Function Documentation

8.39.2.1 getDiscussion_id()

```
Integer capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.getDiscussion↩
_id ()
```

Definition at line 8 of file ResponseDiscussionListDto.java.

8.39.2.2 getQuiz_content()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.getQuiz_↩
content ()
```

Definition at line 20 of file ResponseDiscussionListDto.java.

8.39.2.3 getUser_pdf_name()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.getUser_pdf_↩
name ()
```

Definition at line 14 of file ResponseDiscussionListDto.java.

8.39.2.4 setDiscussion_id()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.setDiscussion_id
(
            Integer discussion_id)
```

Definition at line 11 of file ResponseDiscussionListDto.java.

8.39.2.5 setQuiz_content()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.setQuiz_content
(
            String quiz_content)
```

Definition at line 23 of file ResponseDiscussionListDto.java.

8.39.2.6 setUser_pdf_name()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.setUser_pdf_name
(
            String user_pdf_name)
```

Definition at line 17 of file ResponseDiscussionListDto.java.

8.39.3 Member Data Documentation

8.39.3.1 discussion_id

```
Integer capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.discussion_id
[private]
```

Definition at line 4 of file ResponseDiscussionListDto.java.

8.39.3.2 quiz_content

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.quiz_content
[private]
```

Definition at line 6 of file ResponseDiscussionListDto.java.

8.39.3.3 user_pdf_name

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto.user_pdf_name
[private]
```

Definition at line 5 of file ResponseDiscussionListDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/ResponseDiscussionListDto.java

8.40 capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto Class Reference

Public Member Functions

- Boolean getUse_local_model ()
- void setUse_local_model (Boolean use_local_model)

Private Attributes

- Boolean use_local_model

8.40.1 Detailed Description

Definition at line 3 of file ResponseModelSettingDto.java.

8.40.2 Member Function Documentation

8.40.2.1 getUse_local_model()

```
Boolean capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto.getUse_local_↩
model ()
```

Definition at line 6 of file ResponseModelSettingDto.java.

8.40.2.2 setUse_local_model()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto.setUse_local_model
(
            Boolean use_local_model)
```

Definition at line 10 of file ResponseModelSettingDto.java.

8.40.3 Member Data Documentation

8.40.3.1 use_local_model

```
Boolean capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto.use_local_model
[private]
```

Definition at line 4 of file ResponseModelSettingDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/ResponseModelSettingDto.java

8.41 capstoneProject.Lorenzo.genIz.DTO.response_dto.Response↩ OnlineModelDto Class Reference

Public Member Functions

- String getModel_url ()
- void setModel_url (String model_url)
- String getModel_name ()
- void setModel_name (String model_name)
- String getApi_key ()
- void setApi_key (String api_key)

Private Attributes

- String model_url
- String model_name
- String api_key

8.41.1 Detailed Description

Definition at line 3 of file ResponseOnlineModelDto.java.

8.41.2 Member Function Documentation

8.41.2.1 getApi_key()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.getApi_key ()
```

Definition at line 20 of file ResponseOnlineModelDto.java.

8.41.2.2 getModel_name()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.getModel_name ()
```

Definition at line 14 of file ResponseOnlineModelDto.java.

8.41.2.3 getModel_url()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.getModel_url ()
```

Definition at line 8 of file ResponseOnlineModelDto.java.

8.41.2.4 setApi_key()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.setApi_key (
            String api_key)
```

Definition at line 23 of file ResponseOnlineModelDto.java.

8.41.2.5 setModel_name()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.setModel_name (
            String model_name)
```

Definition at line 17 of file ResponseOnlineModelDto.java.

8.41.2.6 setModel_url()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.setModel_url (
            String model_url)
```

Definition at line 11 of file ResponseOnlineModelDto.java.

8.41.3 Member Data Documentation

8.41.3.1 api_key

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.api_key  [private]
```

Definition at line 6 of file ResponseOnlineModelDto.java.

8.41.3.2 model_name

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.model_name  [private]
```

Definition at line 5 of file ResponseOnlineModelDto.java.

8.41.3.3 model_url

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto.model_url  [private]
```

Definition at line 4 of file ResponseOnlineModelDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/ResponseOnlineModelDto.java

8.42 capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUser↩InfoDto Class Reference

Public Member Functions

- ResponseUserInfoDto (String userPicture, String userName)
- String getUserPicture ()
- void setUserPicture (String userPicture)
- String getUserName ()
- void setUserName (String userName)

Private Attributes

- String userPicture
- String userName

8.42.1 Detailed Description

Definition at line 3 of file ResponseUserInfoDto.java.

8.42.2 Constructor & Destructor Documentation

8.42.2.1 ResponseUserInfoDto()

```
capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.ResponseUserInfoDto (
            String userPicture,
            String userName)
```

Definition at line 7 of file ResponseUserInfoDto.java.

8.42.3 Member Function Documentation

8.42.3.1 getUserName()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.getUserName ()
```

Definition at line 18 of file ResponseUserInfoDto.java.

8.42.3.2 getUserPicture()

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.getUserPicture ()
```

Definition at line 12 of file ResponseUserInfoDto.java.

8.42.3.3 setUserName()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.setUserName (
            String userName)
```

Definition at line 21 of file ResponseUserInfoDto.java.

8.42.3.4 setUserPicture()

```
void capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.setUserPicture (
            String userPicture)
```

Definition at line 15 of file ResponseUserInfoDto.java.

8.42.4 Member Data Documentation

8.42.4.1 userName

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.userName  [private]
```

Definition at line 5 of file ResponseUserInfoDto.java.

8.42.4.2 userPicture

```
String capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto.userPicture  [private]
```

Definition at line 4 of file ResponseUserInfoDto.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/ResponseUserInfoDto.java

8.43 capstoneProject.Lorenzo.genIz.controller.RESTController Class Reference

Public Member Functions

- ResponseUserInfoDto privatePath (OAuth2AuthenticationToken token)

8.43.1 Detailed Description

Definition at line 10 of file RESTController.java.

8.43.2 Member Function Documentation

8.43.2.1 privatePath()

```
ResponseUserInfoDto capstoneProject.Lorenzo.genIz.controller.RESTController.privatePath (
            OAuth2AuthenticationToken token)
```

Definition at line 14 of file RESTController.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/controller/RESTController.java

8.44 capstoneProject.Lorenzo.genIz.security.SecurityConfig Class Reference

Public Member Functions

- SecurityFilterChain securityFilterChain (HttpSecurity http) throws Exception

8.44.1 Detailed Description

Definition at line 15 of file SecurityConfig.java.

8.44.2 Member Function Documentation

8.44.2.1 securityFilterChain()

```
SecurityFilterChain capstoneProject.Lorenzo.genIz.security.SecurityConfig.securityFilterChain
(
            HttpSecurity http) throws Exception
```

Definition at line 18 of file SecurityConfig.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/security/SecurityConfig.java

8.45 capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.TotalTokens Class Reference

Public Member Functions

- Integer getTotal_tokens ()
- void setTotal_tokens (Integer total_tokens)

Private Attributes

- Integer total_tokens

8.45.1 Detailed Description

Definition at line 3 of file TotalTokens.java.

8.45.2 Member Function Documentation

8.45.2.1 getTotal_tokens()

```
Integer capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.TotalTokens.get↩
Total_tokens ()
```

Definition at line 6 of file TotalTokens.java.

8.45.2.2 setTotal_tokens()

```
void capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.TotalTokens.set↩
Total_tokens (
            Integer total_tokens)
```

Definition at line 10 of file TotalTokens.java.

8.45.3 Member Data Documentation

8.45.3.1 total_tokens

```
Integer capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.TotalTokens.↩
total_tokens [private]
```

Definition at line 4 of file TotalTokens.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/map_model_response/TotalTokens.java

8.46 capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest Class Reference

Package Functions

- void testSingletonRowPeristance ()
- void testDeleteAllModelInfo ()
- void testInsertNewApiKeyAndVerifyThatOtherFieldsRemainNull ()
- void verifyThatOtherFieldsDidNotChanged ()

Static Package Attributes

- static MySQLContainer<?> mysql = new MySQLContainer<>("mysql:8.4.0")

Private Attributes

- QuizDaoImpl quizDaoImpl

8.46.1 Detailed Description

Definition at line 20 of file UpdateOnlineModelSettingsTest.java.

8.46.2 Member Function Documentation

8.46.2.1 testDeleteAllModelInfo()

```
void capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest.testDeleteAllModelInfo ()
[package]
```

Definition at line 47 of file UpdateOnlineModelSettingsTest.java.

8.46.2.2 testInsertNewApiKeyAndVerifyThatOtherFieldsRemainNull()

```
void capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest.testInsertNewApiKeyAnd↩
VerifyThatOtherFieldsRemainNull ()  [package]
```

Definition at line 57 of file UpdateOnlineModelSettingsTest.java.

8.46.2.3 testSingletonRowPeristance()

```
void capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest.testSingletonRowPeristance ()
[package]
```

Definition at line 32 of file UpdateOnlineModelSettingsTest.java.

8.46.2.4 verifyThatOtherFieldsDidNotChanged()

```
void capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest.verifyThatOtherFieldsDidNot↩
Changed ()  [package]
```

Definition at line 67 of file UpdateOnlineModelSettingsTest.java.

8.46.3 Member Data Documentation

8.46.3.1 mysql

```
MySQLContainer<?> capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest.mysql = new My↩
SQLContainer<>("mysql:8.4.0")  [static], [package]
```

Definition at line 25 of file UpdateOnlineModelSettingsTest.java.

8.46.3.2 quizDaoImpl

```
QuizDaoImpl capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest.quizDaoImpl [private]
```

Definition at line 28 of file UpdateOnlineModelSettingsTest.java.

The documentation for this class was generated from the following file:

- src/test/java/capstoneProject/Lorenzo/genIz/UpdateOnlineModelSettingsTest.java

8.47 capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO Class Reference

Public Member Functions

- Integer getId ()
- void setId (Integer id)
- String getIdentity_provider_user_id ()
- void setIdentity_provider_user_id (String identity_provider_user_id)
- Boolean getUse_local_model ()
- void setUse_local_model (Boolean use_local_model)

Private Attributes

- Integer id
- String identity_provider_user_id
- Boolean use_local_model

8.47.1 Detailed Description

Definition at line 3 of file UserDataDTO.java.

8.47.2 Member Function Documentation

8.47.2.1 getId()

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.getId ()
```

Definition at line 8 of file UserDataDTO.java.

8.47.2.2 getIdentity_provider_user_id()

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.getIdentity_provider_user_id
()
```

Definition at line 14 of file UserDataDTO.java.

8.47.2.3 getUse_local_model()

```
Boolean capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.getUse_local_model ()
```

Definition at line 20 of file UserDataDTO.java.

8.47.2.4 setId()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.setId (
            Integer id)
```

Definition at line 11 of file UserDataDTO.java.

8.47.2.5 setIdentity_provider_user_id()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.setIdentity_provider_user_id (
            String identity_provider_user_id)
```

Definition at line 17 of file UserDataDTO.java.

8.47.2.6 setUse_local_model()

```
void capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.setUse_local_model (
            Boolean use_local_model)
```

Definition at line 23 of file UserDataDTO.java.

8.47.3 Member Data Documentation

8.47.3.1 id

```
Integer capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.id  [private]
```

Definition at line 4 of file UserDataDTO.java.

8.47.3.2 identity_provider_user_id

```
String capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.identity_provider_user_id
[private]
```

Definition at line 5 of file UserDataDTO.java.

8.47.3.3 use_local_model

```
Boolean capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO.use_local_model  [private]
```

Definition at line 6 of file UserDataDTO.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/DTO/entity_dto/UserDataDTO.java

8.48 capstoneProject.Lorenzo.genIz.entity.UserEntity Class Reference

Public Member Functions

- Integer getId ()
- void setId (Integer id)
- String getIdentity_provider_user_id ()
- void setIdentity_provider_user_id (String identity_provider_user_id)
- List< ChatEntity > getChatEntityList ()
- void setChatEntityList (List< ChatEntity > chatEntityList)
- Boolean getUse_local_model ()
- void setUse_local_model (Boolean use_local_model)
- void addChatEntity (ChatEntity chatEntity)

Private Attributes

- Integer id
- String identity_provider_user_id
- Boolean use_local_model
- List< ChatEntity > chatEntityList

8.48.1 Detailed Description

Definition at line 17 of file UserEntity.java.

8.48.2 Member Function Documentation

8.48.2.1 addChatEntity()

```
void capstoneProject.Lorenzo.genIz.entity.UserEntity.addChatEntity (
            ChatEntity chatEntity)
```

Definition at line 78 of file UserEntity.java.

8.48.2.2 getChatEntityList()

```
List< ChatEntity > capstoneProject.Lorenzo.genIz.entity.UserEntity.getChatEntityList ()
```

Definition at line 57 of file UserEntity.java.

8.48.2.3 getId()

```
Integer capstoneProject.Lorenzo.genIz.entity.UserEntity.getId ()
```

Definition at line 37 of file UserEntity.java.

8.48.2.4 getIdentity_provider_user_id()

```
String capstoneProject.Lorenzo.genIz.entity.UserEntity.getIdentity_provider_user_id ()
```

Definition at line 47 of file UserEntity.java.

8.48.2.5 getUse_local_model()

```
Boolean capstoneProject.Lorenzo.genIz.entity.UserEntity.getUse_local_model ()
```

Definition at line 67 of file UserEntity.java.

8.48.2.6 setChatEntityList()

```
void capstoneProject.Lorenzo.genIz.entity.UserEntity.setChatEntityList (
            List< ChatEntity > chatEntityList)
```

Definition at line 62 of file UserEntity.java.

8.48.2.7 setId()

```
void capstoneProject.Lorenzo.genIz.entity.UserEntity.setId (
            Integer id)
```

Definition at line 42 of file UserEntity.java.

8.48.2.8 setIdentity_provider_user_id()

```
void capstoneProject.Lorenzo.genIz.entity.UserEntity.setIdentity_provider_user_id (
            String identity_provider_user_id)
```

Definition at line 52 of file UserEntity.java.

8.48.2.9 setUse_local_model()

```
void capstoneProject.Lorenzo.genIz.entity.UserEntity.setUse_local_model (
            Boolean use_local_model)
```

Definition at line 72 of file UserEntity.java.

8.48.3 Member Data Documentation

8.48.3.1 chatEntityList

```
List<ChatEntity> capstoneProject.Lorenzo.genIz.entity.UserEntity.chatEntityList  [private]
```

Definition at line 33 of file UserEntity.java.

8.48.3.2 id

```
Integer capstoneProject.Lorenzo.genIz.entity.UserEntity.id  [private]
```

Definition at line 22 of file UserEntity.java.

8.48.3.3 identity_provider_user_id

```
String capstoneProject.Lorenzo.genIz.entity.UserEntity.identity_provider_user_id  [private]
```

Definition at line 25 of file UserEntity.java.

8.48.3.4 use_local_model

```
Boolean capstoneProject.Lorenzo.genIz.entity.UserEntity.use_local_model  [private]
```

Definition at line 28 of file UserEntity.java.

The documentation for this class was generated from the following file:

- src/main/java/capstoneProject/Lorenzo/genIz/entity/UserEntity.java

Chapter 9

File Documentation

9.1 src/main/java/capstoneProject/Lorenzo/genIz/api_format/request/↩ ApiCallRequest.java File Reference

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
```

Classes

- class capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest

Packages

- package capstoneProject.Lorenzo.genIz.api_format.request

9.2 ApiCallRequest.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.api_format.request;
00002
00003 import java.util.ArrayList;
00004 import java.util.HashMap;
00005 import java.util.List;
00006 import java.util.Map;
00007
00008 public class ApiCallRequest {
00009     private String model;
00010     private List<Map<String, String»  messages = new ArrayList<>();
00011
00012     //contructor
00013     public ApiCallRequest(){
00014
00015     }
00016
00017     //getter & setters
00018     public String getModel() {
00019         return model;
00020     }
```

```java

    public void setModel(String model) {
        this.model = model;
    }

    public List<Map<String, String>> getMessages() {
        return messages;
    }

    public void setMessages(List<Map<String, String>> messages) {
        this.messages = messages;
    }

    //function to populate the messages list, each entry is composed of 2 elements, role and content
    public void addField(String role, String content){
        Map<String, String> field = new HashMap<>();
        field.put("role", role);
        field.put("content", content);

        messages.add(field);
    }



}
```

9.3 src/main/java/capstoneProject/Lorenzo/genIz/api_format/request/↩ request_parameters/QuizGenParameter.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.api_format.request.request_parameters.QuizGenParameter

Packages

- package capstoneProject.Lorenzo.genIz.api_format.request.request_parameters

9.4 QuizGenParameter.java

Go to the documentation of this file.

```java
package capstoneProject.Lorenzo.genIz.api_format.request.request_parameters;

public class QuizGenParameter {
    //system prompt, gives the llm instructions on what to do.
    public static final String SYSTEMPROMPT = """
            Role:
            You are a Quiz generator engine, that transform text into insightful quizzes.

            Task:
            You must take the text provided by the user and generate multiple choice quizzes that
            reference the most important infomrations and concepts in the given text for a better learning experience.
            the response should contain:list of quizzes, question, options(from A to D) ,explanation, correct_answer .

            Response Format:
            The response should contain only a json formatted response that follow this standard:
            {
                "quizzes": [
                    {
                        "question": string
                        "options": {
                            "A": string,
                            "B": string,
                            "C": string,
                            "D": string
                        },
                        "explanation": string,
                        "correct_answer": char (the response letter)
```

```
                    }
                ]
            }.

            Example:
            here an example on how the result may look like (the number of questions depend on the provided text):
            {
                "quizzes": [
                    {
                    "question": "What is the primary purpose of a Docker bridge network?",
                    "options": {
                        "A": "To allow containers to communicate on the same host",
                        "B": "To connect containers to a physical hardware router",
                        "C": "To encrypt all traffic between containers",
                        "D": "To replace the need for an IP address"
                    },
                    "explanation": "Bridge networks are the default for standalone containers on a single host.",
                    "answer": "A"
                    },
                    {
                    "question": "Which command is used to pull a Docker image?",
                    "options": {
                        "A": "docker push",
                        "B": "docker pull",
                        "C": "docker fetch",
                        "D": "docker get"
                    },
                    "explanation": "The 'docker pull' command downloads images from a registry like Docker Hub.",
                    "answer": "B"
                    }
                ]
            }

            Constraints:
            1. The response MUST contain ONLY the json response, no introdiction or presentation.
            2. The questions and responses MUST be created by following the material in the provided text.
            3. STRICT CONTEXTUAL ADHERENCE: Do not include information, concepts, or facts that are not explicitly mentioned in the user-provided text. Do not digress from the provided topic or use external training knowledge.
            4. Do NOT put any text outside of the json response.
            5. STRICT FORMAT RULE: do NOT, in any case or situation, wrap the content response with (“`) or any other type of surrounding element.
            """;;


    public static String getSystemprompt() {
        return SYSTEMPROMPT;
    }
}
```

9.5 src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/↩ ApiCallResponse.java File Reference

```
import java.util.List;
import capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.↩
Choices;
import capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.↩
TotalTokens;
```

Classes

- class capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse

Packages

- package capstoneProject.Lorenzo.genIz.api_format.response

9.6 ApiCallResponse.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.api_format.response;

import java.util.List;

import capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Choices;
import capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.TotalTokens;

public class ApiCallResponse {
    private List<Choices> choices;
    private TotalTokens usage;

    public List<Choices> getChoices() {
        return choices;
    }
    public void setChoices(List<Choices> choices) {
        this.choices = choices;
    }
    public TotalTokens getUsage() {
        return usage;
    }
    public void setUsage(TotalTokens usage) {
        this.usage = usage;
    }
}
```

9.7 src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/↩ map_model_response/Choices.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Choices

Packages

- package capstoneProject.Lorenzo.genIz.api_format.response.map_model_response

9.8 Choices.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.api_format.response.map_model_response;

public class Choices {
    private Message message;

    public Message getMessage() {
        return message;
    }

    public void setMessage(Message message) {
        this.message = message;
    }

}
```

9.9 src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/↩ map_model_response/Message.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.Message

Packages

- package capstoneProject.Lorenzo.genIz.api_format.response.map_model_response

9.10 Message.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.api_format.response.map_model_response;

public class Message {
    private String content;

    public String getContent() {
        return content;
    }

    public void setContent(String content) {
        this.content = content;
    }
}
```

9.11 src/main/java/capstoneProject/Lorenzo/genIz/api_format/response/map_model_response/TotalTokens.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.api_format.response.map_model_response.TotalTokens

Packages

- package capstoneProject.Lorenzo.genIz.api_format.response.map_model_response

9.12 TotalTokens.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.api_format.response.map_model_response;

public class TotalTokens {
    private Integer total_tokens;

    public Integer getTotal_tokens() {
        return total_tokens;
    }

    public void setTotal_tokens(Integer total_tokens) {
        this.total_tokens = total_tokens;
    }

}
```

9.13 src/main/java/capstoneProject/Lorenzo/genIz/controller/DataBase↩ Controller.java File Reference

```
import java.util.List;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PatchMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussion↩
ListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSetting↩
Dto;
import capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface;
```

Classes

- class capstoneProject.Lorenzo.genIz.controller.DataBaseController

Packages

- package capstoneProject.Lorenzo.genIz.controller

9.14 DataBaseController.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.controller;
00002
00003 import java.util.List;
00004
00005 import org.springframework.web.bind.annotation.DeleteMapping;
00006 import org.springframework.web.bind.annotation.GetMapping;
00007 import org.springframework.web.bind.annotation.PatchMapping;
00008 import org.springframework.web.bind.annotation.PathVariable;
00009 import org.springframework.web.bind.annotation.PostMapping;
00010 import org.springframework.web.bind.annotation.RequestBody;
00011 import org.springframework.web.bind.annotation.RequestMapping;
00012 import org.springframework.web.bind.annotation.RestController;
00013
00014 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
00015 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
00016 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
00017 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto;
00018 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;
00019 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto;
00020 import capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface;
00021
00022
00023
00024
00025 @RestController
00026 @RequestMapping("/database/interactions")
00027 public class DataBaseController {
```

```java

    //inject the service that implements the dao methods
    ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface;

    public DataBaseController(ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface) {
        this.implementQuizDaoMethodsInterface = implementQuizDaoMethodsInterface;
    }

    //post controller to save a new chat
    @PostMapping("/save-new-chat")
    public ResponseChatListDto saveChat(@RequestBody PostReqChatDto request) {

        ResponseChatListDto savedChat = implementQuizDaoMethodsInterface.saveChat(request);

        return savedChat;
    }

    //get controller to retrieve the list of all available chat fo the specified user
    @GetMapping("/retrieve-all-chats")
    public List<ResponseChatListDto> getAllChats() {

        List<ResponseChatListDto> allUserChats = implementQuizDaoMethodsInterface.retrieveChats();

        return allUserChats;
    }

    //post controller to save new discussion
    @PostMapping("/save-new-discussion")
    public ResponseDiscussionListDto saveDiscussion(@RequestBody PostReqDiscussionDto discussion) {

        ResponseDiscussionListDto saveDiscussion =
implementQuizDaoMethodsInterface.saveDiscussion(discussion);

        return saveDiscussion;
    }

    //get controller to retrieve all the discussions related to the chat
    @GetMapping("/retrieve-all-discussions/{chatId}")
    public List<ResponseDiscussionListDto> getAllDiscussions(@PathVariable Integer chatId) {

        List<ResponseDiscussionListDto> allChatDiscussions =
implementQuizDaoMethodsInterface.retrieveDiscussions(chatId);

        return allChatDiscussions;
    }

    //post controller to save the new model setting
    @PostMapping("/running-model-setting")
    public ResponseModelSettingDto saveModelSetting(@RequestBody PostReqModelSetting modelSetting) {

        ResponseModelSettingDto saveSetting =
implementQuizDaoMethodsInterface.manageModelSettings(modelSetting);

        return saveSetting;
    }

    //get controller to retrieve current model setting
    @GetMapping("/retrieve-model-setting")
    public ResponseModelSettingDto getModelSetting() {

        ResponseModelSettingDto retrieveSetting =
implementQuizDaoMethodsInterface.retrieveModelSettings();

        return retrieveSetting;
    }

    //delete controller for deleting the specified chat
    @DeleteMapping("/delete-chat/{chatId}")
    public void deleteChat(@PathVariable Integer chatId){

        implementQuizDaoMethodsInterface.deleteChat(chatId);
    }

    //patch controller to modify the chat name
    @PatchMapping("/update-chat-name")
    public ResponseChatListDto patchChat(@RequestBody PostReqChatDto postReqChatDto){

        ResponseChatListDto patchedChat = implementQuizDaoMethodsInterface.updateChat(postReqChatDto);

        return patchedChat;
    }

    //delete controller for deleting the specified discussion
    @DeleteMapping("/delete-discussion/{discussionId}")
    public void deleteDiscussion(@PathVariable Integer discussionId){

        implementQuizDaoMethodsInterface.deleteDiscussion(discussionId);
```

```
    }
}
```

9.15 src/main/java/capstoneProject/Lorenzo/genIz/controller/ExportDataController.java File Reference

```
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;
import capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine;
import capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface;
import java.util.List;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
```

Classes

- class capstoneProject.Lorenzo.genIz.controller.ExportDataController

Packages

- package capstoneProject.Lorenzo.genIz.controller

9.16 ExportDataController.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.controller;

import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;
import capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine;
import capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface;

import java.util.List;

import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;


@RestController
@RequestMapping("/export")
public class ExportDataController {

    //inject services to retrieve data from db and export to file
    ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface;
    ExportQuizToLocalMachine exportQuizToLocalMachine;

    public ExportDataController(ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface,
            ExportQuizToLocalMachine exportQuizToLocalMachine) {
        this.implementQuizDaoMethodsInterface = implementQuizDaoMethodsInterface;
        this.exportQuizToLocalMachine = exportQuizToLocalMachine;
    }

    //post controller to pass info and create the local file
    @PostMapping("/createFile/{chatId}")
    public void postMethodName(@PathVariable Integer chatId) {

```

```
        //retrieve the discussion list from the current chat
        List<ResponseDiscussionListDto> discussionListDto = implementQuizDaoMethodsInterface.retrieveDiscussions(chatId);

        //create the file with the retrieved discussions
        List<String> quizList = exportQuizToLocalMachine.formatFileContent(discussionListDto);

        exportQuizToLocalMachine.saveFile(quizList);
    }


}
```

9.17 src/main/java/capstoneProject/Lorenzo/genIz/controller/GenerateQuizController.java File Reference

```
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto;
import capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator;
```

Classes

- class capstoneProject.Lorenzo.genIz.controller.GenerateQuizController

Packages

- package capstoneProject.Lorenzo.genIz.controller

9.18 GenerateQuizController.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.controller;

import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto;
import capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator;


@RestController
@RequestMapping("/generate-quizzes")
public class GenerateQuizController {

    //injecting services to perform Api calls
    LocalModelApiFlowAggregator localModelApiFlowAggregator;

    public GenerateQuizController(LocalModelApiFlowAggregator localModelApiFlowAggregator) {
        this.localModelApiFlowAggregator = localModelApiFlowAggregator;
    }

    @PostMapping("/generate")
    public ResponseDataDto generateQuizzesWithLocalModel(@RequestBody PostReqUserTextDto userTextDto) {
```

```

        ResponseDataDto apiCallResponse = localModelApiFlowAggregator.performApiCall(userTextDto);

        ResponseDataDto validateApiResponse =
    localModelApiFlowAggregator.validateApiCallResult(apiCallResponse);

        return validateApiResponse;
    }


}
```

9.19 src/main/java/capstoneProject/Lorenzo/genIz/controller/OnlineModelController.java File Reference

```
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface;
```

Classes

- class capstoneProject.Lorenzo.genIz.controller.OnlineModelController

Packages

- package capstoneProject.Lorenzo.genIz.controller

9.20 OnlineModelController.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.controller;

import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface;

@RestController
@RequestMapping("/online-model/settings")
public class OnlineModelController {

    //inject database methods
    ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface;

    public OnlineModelController(ImplementQuizDaoMethodsInterface implementQuizDaoMethodsInterface){
        this.implementQuizDaoMethodsInterface = implementQuizDaoMethodsInterface;
    }

    //post model to save new model settings
    @PostMapping("/new-settings")
    public void saveNewSettings(@RequestBody PostReqOnlineModelDto postReqOnlineModelDto){
        implementQuizDaoMethodsInterface.saveNewOnlineModelSettings(postReqOnlineModelDto);
    }

    //delete method to remove current model settings
    @DeleteMapping("/delete-settings")
    public void deleteSettings(){
        implementQuizDaoMethodsInterface.deleteOnlineModelSettings();
    }
}
```

9.21 src/main/java/capstoneProject/Lorenzo/genIz/controller/↩ RESTController.java File Reference

```
import org.springframework.security.oauth2.client.authentication.OAuth2↩
AuthenticationToken;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto;
```

Classes

- class capstoneProject.Lorenzo.genIz.controller.RESTController

Packages

- package capstoneProject.Lorenzo.genIz.controller

9.22 RESTController.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.controller;
00002
00003 import org.springframework.security.oauth2.client.authentication.OAuth2AuthenticationToken;
00004 import org.springframework.web.bind.annotation.GetMapping;
00005 import org.springframework.web.bind.annotation.RestController;
00006
00007 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto;
00008
00009 @RestController
00010 public class RESTController {
00011
00012     //retrieving user info to populate user details in frontend
00013     @GetMapping("/user-info")
00014     public ResponseUserInfoDto privatePath(OAuth2AuthenticationToken token) {
00015
00016         String userPicture = token.getPrincipal().getAttribute("picture");
00017         String userName = token.getPrincipal().getAttribute("name");
00018
00019         ResponseUserInfoDto userInfoDto = new ResponseUserInfoDto(userPicture, userName);
00020
00021         return userInfoDto;
00022     }
00023
00024 }
```

9.23 src/main/java/capstoneProject/Lorenzo/genIz/DAO/QuizDao↩ Impl.java File Reference

```
import java.util.ArrayList;
import java.util.List;
import org.springframework.security.access.AccessDeniedException;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.oauth2.client.authentication.OAuth2↩
AuthenticationToken;
import org.springframework.stereotype.Repository;
```

```
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModel↩
Dto;
import capstoneProject.Lorenzo.genIz.entity.ChatEntity;
import capstoneProject.Lorenzo.genIz.entity.DiscussionEntity;
import capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity;
import capstoneProject.Lorenzo.genIz.entity.UserEntity;
import jakarta.persistence.EntityManager;
import jakarta.persistence.NoResultException;
import jakarta.persistence.TypedQuery;
import jakarta.transaction.Transactional;
```

Classes

- class capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl

Packages

- package capstoneProject.Lorenzo.genIz.DAO

9.24 QuizDaoImpl.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.DAO;
00002
00003 import java.util.ArrayList;
00004 import java.util.List;
00005
00006 import org.springframework.security.access.AccessDeniedException;
00007 import org.springframework.security.core.context.SecurityContextHolder;
00008 import org.springframework.security.oauth2.client.authentication.OAuth2AuthenticationToken;
00009 import org.springframework.stereotype.Repository;
00010
00011 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto;
00012 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto;
00013 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO;
00014 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
00015 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
00016 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
00017 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
00018 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;
00019 import capstoneProject.Lorenzo.genIz.entity.ChatEntity;
00020 import capstoneProject.Lorenzo.genIz.entity.DiscussionEntity;
00021 import capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity;
00022 import capstoneProject.Lorenzo.genIz.entity.UserEntity;
00023 import jakarta.persistence.EntityManager;
00024 import jakarta.persistence.NoResultException;
00025 import jakarta.persistence.TypedQuery;
00026 import jakarta.transaction.Transactional;
00027
00028 @Repository
00029 public class QuizDaoImpl implements QuizDaoInterface{
00030
00031     //inject entity manager to interact with DB
00032     EntityManager entityManager;
00033
00034     public QuizDaoImpl(EntityManager entityManager) {
00035         this.entityManager = entityManager;
```

```
    }

    //managing user login, if the user already exist, populate the DTO with user data
    //if the user does not exist, create a new user
    @Override
    @Transactional
    public UserDataDTO manageUser(OAuth2AuthenticationToken token) {

        String uniqueUserIdentifier = token.getPrincipal().getAttribute("sub");

        UserDataDTO userInfoDto = new UserDataDTO();

        try{
            //try query to identify if the user already exist
            TypedQuery<UserEntity> currentUserQuery = entityManager.createQuery(
                "SELECT u FROM UserEntity u WHERE u.identity_provider_user_id = :uniqueId", UserEntity.class);

            currentUserQuery.setParameter("uniqueId", uniqueUserIdentifier);

            UserEntity currentUser = currentUserQuery.getSingleResult();

            userInfoDto.setId(currentUser.getId());
            userInfoDto.setIdentity_provider_user_id(currentUser.getIdentity_provider_user_id());
            userInfoDto.setUse_local_model(currentUser.getUse_local_model());

            return userInfoDto;

        } catch (NoResultException e){
            //if the user does not exixt create a new one with the OAuth2 provided info
            UserEntity tempNewUser = new UserEntity();
            tempNewUser.setIdentity_provider_user_id(uniqueUserIdentifier);
            tempNewUser.setUse_local_model(true);

            UserEntity newUser = entityManager.merge(tempNewUser);

            userInfoDto.setId(newUser.getId());
            userInfoDto.setIdentity_provider_user_id(newUser.getIdentity_provider_user_id());
            userInfoDto.setUse_local_model(newUser.getUse_local_model());

            return userInfoDto;

        }
    }

    //create a new chat based on the user id and the chat name
    @Override
    @Transactional
    public ChatDataDto saveChat(UserDataDTO userDataDTO, PostReqChatDto postReqChatDto) {

        //find the user based on the userDto
        UserEntity currentUser = entityManager.find(UserEntity.class, userDataDTO.getId());
        if(currentUser == null){
            throw new IllegalArgumentException("no user found");
        }

        //create the dto for data transfer
        ChatDataDto chatInfoDto = new ChatDataDto();

        //create the chat entity
        ChatEntity tempNewChat = new ChatEntity();
        tempNewChat.setChat_name(postReqChatDto.getChatName());

        //create the relation between chat and user entity
        tempNewChat.setDefUserEntity(currentUser);
        // currentUser.addChatEntity(tempNewChat);

        //save the new chat
        ChatEntity newChatEntity = entityManager.merge(tempNewChat);

        //map the dto and return the new data
        chatInfoDto.setChat_id(newChatEntity.getChat_id());
        chatInfoDto.setChat_name(newChatEntity.getChat_name());
        chatInfoDto.setUser_id(currentUser.getId());

        return chatInfoDto;
    }

    //retrieve all the chats owned by the specific users
    @Override
    public List<ChatDataDto> retrieveChats(UserDataDTO userDataDTO) {

        //find the user based on the userDto
        UserEntity currentUser = entityManager.find(UserEntity.class, userDataDTO.getId());
        if(currentUser == null){
            throw new NoResultException("no user found");
        }
```

```java

        //query the db to return only chats that are owned by the specific user

        TypedQuery<ChatEntity> chatQuery = entityManager.createQuery(
            "SELECT c FROM ChatEntity c WHERE c.defUserEntity.id = :userId ORDER BY c.chat_id DESC", ChatEntity.class);

        chatQuery.setParameter("userId", userDataDTO.getId());

        //retrieve the list of chats
        List<ChatEntity> userChats = chatQuery.getResultList();

        //map the chatEntity to a list of dto
        List<ChatDataDto> listUserChatsDto = new ArrayList<>();
        for(ChatEntity chat : userChats){
            ChatDataDto tempChatDataDto = new ChatDataDto();
            tempChatDataDto.setChat_id(chat.getChat_id());
            tempChatDataDto.setChat_name(chat.getChat_name());
            tempChatDataDto.setUser_id(userDataDTO.getId());
            listUserChatsDto.add(tempChatDataDto);
        }

        return listUserChatsDto;

    }

    //save the current generated quiz discussion
    @Override
    @Transactional
    public DiscussionDataDto saveDiscussion(PostReqDiscussionDto postReqDiscussionDto) {

        //retrieve the chat where to save the discussion
        ChatEntity currentChat = entityManager.find(ChatEntity.class, postReqDiscussionDto.getChat_id());

        if(currentChat == null){
            throw new NoResultException("no chat found");
        }

        //retrieve the current user unique identifier from spring security
        String sub = SecurityContextHolder.getContext().getAuthentication().getName();

        //compare the retrieved user with the current one, to block requests from other authenticated users
        if(currentChat.getDefUserEntity().getIdentity_provider_user_id().equals(sub)){

            //create a new discussion
            DiscussionEntity tempDiscussionEntity = new DiscussionEntity();

            tempDiscussionEntity.setUser_pdf_name(postReqDiscussionDto.getUser_pdf_name());
            tempDiscussionEntity.setQuiz_content(postReqDiscussionDto.getQuiz_content());

            //create the relation between the cat and the discussion
            // currentChat.addDiscussion(tempDiscussionEntity);
            tempDiscussionEntity.setDefChatEntity(currentChat);

            //save the new discussion
            DiscussionEntity currentDiscussion = entityManager.merge(tempDiscussionEntity);

            //create the dto and return it
            DiscussionDataDto currentDiscussionDataDto = new DiscussionDataDto();
            currentDiscussionDataDto.setDiscussion_id(currentDiscussion.getDiscussion_id());
            currentDiscussionDataDto.setUser_pdf_name(currentDiscussion.getUser_pdf_name());
            currentDiscussionDataDto.setQuiz_content(currentDiscussion.getQuiz_content());
            currentDiscussionDataDto.setChat_id(postReqDiscussionDto.getChat_id());

            return currentDiscussionDataDto;

        } else {
            throw new AccessDeniedException("user not authorized to access this chat");
        }

    }

    //retrieve all the discussions of a chat
    @Override
    public List<DiscussionDataDto> retrieveDiscussions(Integer chatId) {

        //retrieve the chat that holds the requested conversations
        ChatEntity currentChat = entityManager.find(ChatEntity.class, chatId);

        if(currentChat == null){
            throw new NoResultException("no chat found");
        }

        //retrieve the current user unique identifier from spring security
        String sub = SecurityContextHolder.getContext().getAuthentication().getName();
```

```java

        //compare the retrieved user with the current one, to block requests from other authenticated users
        if(currentChat.getDefUserEntity().getIdentity_provider_user_id().equals(sub)){

            //retrieve all the discussions related to the chat
            TypedQuery<DiscussionEntity> discussionQuery = entityManager.createQuery(
                "SELECT d FROM DiscussionEntity d WHERE d.defChatEntity.chat_id = :currentChatId", DiscussionEntity.class);

            discussionQuery.setParameter("currentChatId", chatId);

            //query the db
            List<DiscussionEntity> chatDiscussions = discussionQuery.getResultList();

            //create the discussionDto list
            List<DiscussionDataDto> listChatDiscussionDto = new ArrayList<>();
            for(DiscussionEntity discussion : chatDiscussions){
                DiscussionDataDto discussionDataDto = new DiscussionDataDto();
                discussionDataDto.setDiscussion_id(discussion.getDiscussion_id());
                discussionDataDto.setUser_pdf_name(discussion.getUser_pdf_name());
                discussionDataDto.setQuiz_content(discussion.getQuiz_content());
                discussionDataDto.setChat_id(chatId);

                listChatDiscussionDto.add(discussionDataDto);
            }

            return listChatDiscussionDto;

        } else {
            throw new AccessDeniedException("user not authorized to access this chat");
        }

    }

    //toggle the switch to tell the application to use the local model or the specified Internet one
    @Override
    @Transactional
    public UserDataDTO manageModelSettings(UserDataDTO userDataDTO, PostReqModelSetting postReqModelSetting) {

        //retrieve the current user
        UserEntity currentUser = entityManager.find(UserEntity.class, userDataDTO.getId());

        //if the request is acutally changing the setting, do the modification, else do nothing
        if(!currentUser.getUse_local_model().equals(postReqModelSetting.getModelSetting())){
            currentUser.setUse_local_model(postReqModelSetting.getModelSetting());
        }

        //save the new user and map it with the new dto
        currentUser = entityManager.merge(currentUser);

        UserDataDTO newUserDataDTO = new UserDataDTO();
        newUserDataDTO.setId(currentUser.getId());
        newUserDataDTO.setIdentity_provider_user_id(currentUser.getIdentity_provider_user_id());
        newUserDataDTO.setUse_local_model(currentUser.getUse_local_model());

        return newUserDataDTO;
    }

    //retrieve the current model setting
    @Override
    public UserDataDTO retrieveModelSettings(UserDataDTO userDataDTO) {

        //retrieve the current user info
        UserEntity currentUser = entityManager.find(UserEntity.class, userDataDTO.getId());

        //map the user dto
        UserDataDTO newUserDataDTO = new UserDataDTO();
        newUserDataDTO.setId(currentUser.getId());
        newUserDataDTO.setIdentity_provider_user_id(currentUser.getIdentity_provider_user_id());
        newUserDataDTO.setUse_local_model(currentUser.getUse_local_model());

        return newUserDataDTO;
    }

    //delete the specified chat
    @Override
    @Transactional
    public void deleteChat(Integer chatId) {

        //retrieve the chat object
        ChatEntity currentChat = entityManager.find(ChatEntity.class, chatId);

        //retrieve the current user unique identifier from spring security
        String sub = SecurityContextHolder.getContext().getAuthentication().getName();

```

```
        //compare the retrieved user with the current one, to block requests from other authenticated users
        if(currentChat.getDefUserEntity().getIdentity_provider_user_id().equals(sub)){

            //delete the chat entity from the db
            entityManager.remove(currentChat);

        } else {
            throw new AccessDeniedException("user not authorized to access this chat");
        }

    }

    //update chatname
    @Override
    @Transactional
    public ChatDataDto updateChat(PostReqChatDto postReqChatDto) {

        //retrieve the current chat
        ChatEntity currentChat = entityManager.find(ChatEntity.class, postReqChatDto.getChat_id());

        //retrieve the current user unique identifier from spring security
        String sub = SecurityContextHolder.getContext().getAuthentication().getName();

        //compare the retrieved user with the current one, to block requests from other authenticated users
        if(currentChat.getDefUserEntity().getIdentity_provider_user_id().equals(sub)){

            //set the new value of chatName and let Jpa identify the change and apply it
            currentChat.setChat_name(postReqChatDto.getChatName());

            //return the new chat dto
            ChatDataDto updatedChat = new ChatDataDto();
            updatedChat.setChat_id(currentChat.getChat_id());
            updatedChat.setChat_name(currentChat.getChat_name());

            return updatedChat;

        } else {
            throw new AccessDeniedException("user not authorized to access this chat");
        }

    }

    //delete the specified discussion
    @Override
    @Transactional
    public void deleteDiscussion(Integer discussionId) {

        //retrieve current discussion
        DiscussionEntity currentDiscussion = entityManager.find(DiscussionEntity.class, discussionId);

        //retrieve the current user unique identifier from spring security
        String sub = SecurityContextHolder.getContext().getAuthentication().getName();

        //compare the retrieved user with the current one, to block requests from other authenticated users

        if(currentDiscussion.getDefChatEntity().getDefUserEntity().getIdentity_provider_user_id().equals(sub)){

            //delete the discussion
            entityManager.remove(currentDiscussion);

        } else {
            throw new AccessDeniedException("user not authorized to access this discussion");
        }

    }

    @Override
    @Transactional
    public void saveNewOnlineModelSettings(PostReqOnlineModelDto postReqOnlineModelDto) {

        //no control over current user because online model is shared across all users on the same machine

        //retrieve current values
        OnlineModelEntity currentValues = entityManager.find(OnlineModelEntity.class, 1);
        if(currentValues == null){
            currentValues = new OnlineModelEntity();
            currentValues.setModelUrl("null");
            currentValues.setModelName("null");
            currentValues.setApiKey("null");
        }

        //check what has been modified and update it
        if(postReqOnlineModelDto.getModel_url() != null &&
```

```
!postReqOnlineModelDto.getModel_url().isEmpty()){
            currentValues.setModelUrl(postReqOnlineModelDto.getModel_url());
        }

        if(postReqOnlineModelDto.getModel_name() != null &&
!postReqOnlineModelDto.getModel_name().isEmpty()){
            currentValues.setModelName(postReqOnlineModelDto.getModel_name());
        }

        if(postReqOnlineModelDto.getApi_key() != null &&
!postReqOnlineModelDto.getApi_key().isEmpty()){
            currentValues.setApiKey(postReqOnlineModelDto.getApi_key());
        }

        //merge the changes
        entityManager.merge(currentValues);
    }

    @Override
    public ResponseOnlineModelDto retrieveOnlineModelSettings() {

        //no control over current user because online model is shared across all users on the same
machine

        //retrieve current model settings
        OnlineModelEntity currentValues = entityManager.find(OnlineModelEntity.class, 1);
        if(currentValues == null){
            currentValues = new OnlineModelEntity();
            currentValues.setModelUrl("null");
            currentValues.setModelName("null");
            currentValues.setApiKey("null");
        }

        //map the DTO
        ResponseOnlineModelDto responseOnlineModelDto = new ResponseOnlineModelDto();
        responseOnlineModelDto.setModel_url(currentValues.getModelUrl());
        responseOnlineModelDto.setModel_name(currentValues.getModelName());
        responseOnlineModelDto.setApi_key(currentValues.getApiKey());

        return responseOnlineModelDto;
    }

    @Override
    @Transactional
    public void deleteOnlineModelSettings() {

        //no control over current user because online model is shared across all users on the same
machine

        //retrieve current model settings
        OnlineModelEntity currentValues = entityManager.find(OnlineModelEntity.class, 1);

        //delete current model settings
        if (currentValues != null) {
            currentValues.setModelUrl("null");
            currentValues.setModelName("null");
            currentValues.setApiKey("null");
            entityManager.merge(currentValues);
        }
    }

}
```

9.25 src/main/java/capstoneProject/Lorenzo/genIz/DAO/QuizDao↩ Interface.java File Reference

```
import java.util.List;
import org.springframework.security.oauth2.client.authentication.OAuth2↩
AuthenticationToken;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
```

```
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModel↩
Dto;
```

Classes

- interface capstoneProject.Lorenzo.genIz.DAO.QuizDaoInterface

Packages

- package capstoneProject.Lorenzo.genIz.DAO

9.26 QuizDaoInterface.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.DAO;
00002
00003 import java.util.List;
00004
00005 import org.springframework.security.oauth2.client.authentication.OAuth2AuthenticationToken;
00006
00007 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto;
00008 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto;
00009 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO;
00010 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
00011 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
00012 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
00013 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
00014 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;
00015
00016 public interface QuizDaoInterface {
00017
00018     public UserDataDTO manageUser(OAuth2AuthenticationToken token);
00019
00020     public ChatDataDto saveChat(UserDataDTO userDataDTO, PostReqChatDto postReqChatDto);
00021
00022     public List<ChatDataDto> retrieveChats(UserDataDTO userDataDTO);
00023
00024     public DiscussionDataDto saveDiscussion(PostReqDiscussionDto postReqDiscussionDto);
00025
00026     public List<DiscussionDataDto> retrieveDiscussions(Integer chatId);
00027
00028     public UserDataDTO manageModelSettings(UserDataDTO userDataDTO, PostReqModelSetting
      postReqModelSetting);
00029
00030     public UserDataDTO retrieveModelSettings(UserDataDTO userDataDTO);
00031
00032     public void deleteChat(Integer chatId);
00033
00034     public ChatDataDto updateChat(PostReqChatDto postReqChatDto);
00035
00036     public void deleteDiscussion(Integer discussionId);
00037
00038     public void saveNewOnlineModelSettings(PostReqOnlineModelDto postReqOnlineModelDto);
00039
00040     public ResponseOnlineModelDto retrieveOnlineModelSettings();
00041
00042     public void deleteOnlineModelSettings();
00043 }
```

9.27 src/main/java/capstoneProject/Lorenzo/genIz/DTO/entity_dto/Chat↩ DataDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.entity_dto

9.28 ChatDataDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.entity_dto;

public class ChatDataDto {
    private Integer chat_id;
    private String chat_name;
    private Integer user_id;

    public Integer getChat_id() {
        return chat_id;
    }
    public void setChat_id(Integer chat_id) {
        this.chat_id = chat_id;
    }
    public String getChat_name() {
        return chat_name;
    }
    public void setChat_name(String chat_name) {
        this.chat_name = chat_name;
    }
    public Integer getUser_id() {
        return user_id;
    }
    public void setUser_id(Integer user_id) {
        this.user_id = user_id;
    }


}
```

9.29 src/main/java/capstoneProject/Lorenzo/genIz/DTO/entity_dto/↩ DiscussionDataDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.entity_dto

9.30 DiscussionDataDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.entity_dto;

public class DiscussionDataDto {
    private Integer discussion_id;
    private String user_pdf_name;
    private String quiz_content;
    private Integer chat_id;

    public Integer getDiscussion_id() {
        return discussion_id;
    }
```

```
    public void setDiscussion_id(Integer discussion_id) {
        this.discussion_id = discussion_id;
    }
    public String getUser_pdf_name() {
        return user_pdf_name;
    }
    public void setUser_pdf_name(String user_pdf_name) {
        this.user_pdf_name = user_pdf_name;
    }
    public String getQuiz_content() {
        return quiz_content;
    }
    public void setQuiz_content(String quiz_content) {
        this.quiz_content = quiz_content;
    }
    public Integer getChat_id() {
        return chat_id;
    }
    public void setChat_id(Integer chat_id) {
        this.chat_id = chat_id;
    }


}
```

9.31 src/main/java/capstoneProject/Lorenzo/genIz/DTO/entity_dto/User↩ DataDTO.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO

Packages

- package capstoneProject.Lorenzo.genIz.DTO.entity_dto

9.32 UserDataDTO.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.entity_dto;

public class UserDataDTO {
    private Integer id;
    private String identity_provider_user_id;
    private Boolean use_local_model;

    public Integer getId() {
        return id;
    }
    public void setId(Integer id) {
        this.id = id;
    }
    public String getIdentity_provider_user_id() {
        return identity_provider_user_id;
    }
    public void setIdentity_provider_user_id(String identity_provider_user_id) {
        this.identity_provider_user_id = identity_provider_user_id;
    }
    public Boolean getUse_local_model() {
        return use_local_model;
    }
    public void setUse_local_model(Boolean use_local_model) {
        this.use_local_model = use_local_model;
    }


}
```

9.33 src/main/java/capstoneProject/Lorenzo/genIz/DTO/ErrorResponse↩ Dto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO

9.34 ErrorResponseDto.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.DTO;
00002
00003 public class ErrorResponseDto {
00004     public String exceptionErrorMessage;
00005
00006     public ErrorResponseDto(String exceptionErrorMessage) {
00007         this.exceptionErrorMessage = exceptionErrorMessage;
00008     }
00009
00010     public String getExceptionErrorMessage() {
00011         return exceptionErrorMessage;
00012     }
00013
00014     public void setExceptionErrorMessage(String exceptionErrorMessage) {
00015         this.exceptionErrorMessage = exceptionErrorMessage;
00016     }
00017
00018
00019 }
```

9.35 src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/↩ PostReqChatDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.request_dto

9.36 PostReqChatDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.request_dto;

public class PostReqChatDto {
    private String chatName;
    private Integer chat_id;

    public String getChatName() {
        return chatName;
    }

    public void setChatName(String chatName) {
        this.chatName = chatName;
    }

    public Integer getChat_id() {
        return chat_id;
    }

    public void setChat_id(Integer chat_id) {
        this.chat_id = chat_id;
    }

}
```

9.37 src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/↩ PostReqDiscussionDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.request_dto

9.38 PostReqDiscussionDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.request_dto;

public class PostReqDiscussionDto {
    private Integer chat_id;
    private String user_pdf_name;
    private String quiz_content;

    public Integer getChat_id() {
        return chat_id;
    }
    public void setChat_id(Integer chat_id) {
        this.chat_id = chat_id;
    }
    public String getUser_pdf_name() {
        return user_pdf_name;
    }
    public void setUser_pdf_name(String user_pdf_name) {
        this.user_pdf_name = user_pdf_name;
    }
    public String getQuiz_content() {
        return quiz_content;
    }
    public void setQuiz_content(String quiz_content) {
        this.quiz_content = quiz_content;
    }



}
```

9.39 src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/↩ PostReqModelSetting.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting

Packages

- package capstoneProject.Lorenzo.genIz.DTO.request_dto

9.40 PostReqModelSetting.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.DTO.request_dto;
00002
00003 public class PostReqModelSetting {
00004     private Boolean modelSetting;
00005
00006     public Boolean getModelSetting() {
00007         return modelSetting;
00008     }
00009
00010     public void setModelSetting(Boolean modelSetting) {
00011         this.modelSetting = modelSetting;
00012     }
00013
00014
00015 }
```

9.41 src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/↩ PostReqOnlineModelDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.request_dto

9.42 PostReqOnlineModelDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.request_dto;

public class PostReqOnlineModelDto {
    private String model_url;
    private String model_name;
    private String api_key;

    public String getModel_url() {
        return model_url;
    }
    public void setModel_url(String model_url) {
        this.model_url = model_url;
    }
    public String getModel_name() {
        return model_name;
    }
    public void setModel_name(String model_name) {
        this.model_name = model_name;
    }
    public String getApi_key() {
        return api_key;
    }
    public void setApi_key(String api_key) {
        this.api_key = api_key;
    }
}
```

9.43 src/main/java/capstoneProject/Lorenzo/genIz/DTO/request_dto/↩ PostReqUserTextDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.request_dto

9.44 PostReqUserTextDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.request_dto;

public class PostReqUserTextDto {
    private String userText;
    private Boolean useLocalModel;

    public String getUserText() {
        return userText;
    }

    public void setUserText(String userText) {
        this.userText = userText;
    }

    public Boolean getUseLocalModel() {
        return useLocalModel;
    }

    public void setUseLocalModel(Boolean useLocalModel) {
        this.useLocalModel = useLocalModel;
    }


}
```

9.45 src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/↩ ResponseChatListDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.response_dto

9.46 ResponseChatListDto.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.DTO.response_dto;
00002
00003 public class ResponseChatListDto {
00004     private Integer chat_id;
00005     private String chat_name;
00006
00007     public Integer getChat_id() {
00008         return chat_id;
00009     }
00010     public void setChat_id(Integer chat_id) {
00011         this.chat_id = chat_id;
00012     }
00013     public String getChat_name() {
00014         return chat_name;
00015     }
00016     public void setChat_name(String chat_name) {
00017         this.chat_name = chat_name;
00018     }
00019
00020 }
```

9.47 src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/↩ ResponseDiscussionListDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.response_dto

9.48 ResponseDiscussionListDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.response_dto;

public class ResponseDiscussionListDto {
    private Integer discussion_id;
    private String user_pdf_name;
    private String quiz_content;

    public Integer getDiscussion_id() {
        return discussion_id;
    }
    public void setDiscussion_id(Integer discussion_id) {
        this.discussion_id = discussion_id;
    }
    public String getUser_pdf_name() {
        return user_pdf_name;
    }
    public void setUser_pdf_name(String user_pdf_name) {
        this.user_pdf_name = user_pdf_name;
    }
    public String getQuiz_content() {
        return quiz_content;
    }
    public void setQuiz_content(String quiz_content) {
        this.quiz_content = quiz_content;
    }

}
```

9.49 src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/↩ ResponseModelSettingDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.response_dto

9.50 ResponseModelSettingDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.response_dto;

public class ResponseModelSettingDto {
    private Boolean use_local_model;

    public Boolean getUse_local_model() {
        return use_local_model;
    }

    public void setUse_local_model(Boolean use_local_model) {
        this.use_local_model = use_local_model;
    }

}
```

9.51 src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/↩ ResponseOnlineModelDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.response_dto

9.52 ResponseOnlineModelDto.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.DTO.response_dto;
00002
00003 public class ResponseOnlineModelDto {
00004     private String model_url;
00005     private String model_name;
00006     private String api_key;
00007
00008     public String getModel_url() {
00009         return model_url;
00010     }
00011     public void setModel_url(String model_url) {
00012         this.model_url = model_url;
00013     }
00014     public String getModel_name() {
00015         return model_name;
00016     }
00017     public void setModel_name(String model_name) {
00018         this.model_name = model_name;
00019     }
00020     public String getApi_key() {
00021         return api_key;
00022     }
00023     public void setApi_key(String api_key) {
00024         this.api_key = api_key;
00025     }
00026 }
```

9.53 src/main/java/capstoneProject/Lorenzo/genIz/DTO/response_dto/↩ ResponseUserInfoDto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseUserInfoDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO.response_dto

9.54 ResponseUserInfoDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO.response_dto;

public class ResponseUserInfoDto {
    private String userPicture;
    private String userName;

    public ResponseUserInfoDto(String userPicture, String userName) {
        this.userPicture = userPicture;
        this.userName = userName;
    }

    public String getUserPicture() {
        return userPicture;
    }
    public void setUserPicture(String userPicture) {
        this.userPicture = userPicture;
    }
    public String getUserName() {
        return userName;
    }
    public void setUserName(String userName) {
        this.userName = userName;
    }


}
```

9.55 src/main/java/capstoneProject/Lorenzo/genIz/DTO/ResponseData↩ Dto.java File Reference

Classes

- class capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto

Packages

- package capstoneProject.Lorenzo.genIz.DTO

9.56 ResponseDataDto.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.DTO;

public class ResponseDataDto {
    private String llmOutput;
    private Integer tokenCount;

    public ResponseDataDto(String llmOutput, Integer tokenCount) {
        this.llmOutput = llmOutput;
        this.tokenCount = tokenCount;
    }

    public String getLlmOutput() {
        return llmOutput;
    }

    public void setLlmOutput(String llmOutput) {
        this.llmOutput = llmOutput;
    }

    public Integer getTokenCount() {
        return tokenCount;
    }

    public void setTokenCount(Integer tokenCount) {
        this.tokenCount = tokenCount;
    }

}
```

9.57 src/main/java/capstoneProject/Lorenzo/genIz/entity/ChatEntity.java File Reference

```
import java.util.ArrayList;
import java.util.List;
import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.JoinColumn;
import jakarta.persistence.ManyToOne;
import jakarta.persistence.OneToMany;
import jakarta.persistence.Table;
```

Classes

- class capstoneProject.Lorenzo.genIz.entity.ChatEntity

Packages

- package capstoneProject.Lorenzo.genIz.entity

9.58 ChatEntity.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.entity;
00002
00003 import java.util.ArrayList;
00004 import java.util.List;
00005
00006 import jakarta.persistence.CascadeType;
00007 import jakarta.persistence.Column;
00008 import jakarta.persistence.Entity;
00009 import jakarta.persistence.GeneratedValue;
00010 import jakarta.persistence.GenerationType;
00011 import jakarta.persistence.Id;
00012 import jakarta.persistence.JoinColumn;
00013 import jakarta.persistence.ManyToOne;
00014 import jakarta.persistence.OneToMany;
00015 import jakarta.persistence.Table;
00016
00017 @Entity
00018 @Table(name="chats")
00019 public class ChatEntity {
00020
00021     @Id
00022     @GeneratedValue(strategy=GenerationType.IDENTITY)
00023     @Column(name="chat_id")
00024     private Integer chat_id;
00025
00026     @Column(name="chat_name")
00027     private String chat_name;
00028
00029     //each chat has only one owner (user)
00030     //this is the owning side
00031     @ManyToOne
00032     @JoinColumn(name="user_id")
00033     private UserEntity defUserEntity;
00034
00035     //list of all the discussions for the spefic chat
00036     //this is the owned side
```

```
    @OneToMany(mappedBy="defChatEntity", cascade=CascadeType.ALL)
    private List<DiscussionEntity> discussionEntityList;

    public Integer getChat_id() {
        return chat_id;
    }

    public void setChat_id(Integer chat_id) {
        this.chat_id = chat_id;
    }

    public String getChat_name() {
        return chat_name;
    }

    public void setChat_name(String chat_name) {
        this.chat_name = chat_name;
    }

    public UserEntity getDefUserEntity() {
        return defUserEntity;
    }

    public void setDefUserEntity(UserEntity defUserEntity) {
        this.defUserEntity = defUserEntity;
    }

    public List<DiscussionEntity> getDiscussionEntity() {
        return discussionEntityList;
    }

    public void setDiscussionEntity(List<DiscussionEntity> discussionEntity) {
        this.discussionEntityList = discussionEntity;
    }

    //method to add a discussion to a chat when saved
    public void addDiscussion(DiscussionEntity discussionEntity){
        if(discussionEntityList == null){
            discussionEntityList = new ArrayList<>();
        }

        discussionEntityList.add(discussionEntity);
    }
}
```

9.59 src/main/java/capstoneProject/Lorenzo/genlz/entity/Discussion↩ Entity.java File Reference

```
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.JoinColumn;
import jakarta.persistence.ManyToOne;
import jakarta.persistence.Table;
```

Classes

- class capstoneProject.Lorenzo.genlz.entity.DiscussionEntity

Packages

- package capstoneProject.Lorenzo.genlz.entity

9.60 DiscussionEntity.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.entity;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.JoinColumn;
import jakarta.persistence.ManyToOne;
import jakarta.persistence.Table;

@Entity
@Table(name="discussions")
public class DiscussionEntity {

    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="discussion_id")
    private Integer discussion_id;

    @Column(name="user_pdf_name")
    private String user_pdf_name;

    @Column(name="quiz_content")
    private String quiz_content;

    //each discussion belong to a single chat
    //this is the owning side
    @ManyToOne
    @JoinColumn(name="chat_id")
    private ChatEntity defChatEntity;

    public Integer getDiscussion_id() {
        return discussion_id;
    }

    public void setDiscussion_id(Integer discussion_id) {
        this.discussion_id = discussion_id;
    }

    public String getUser_pdf_name() {
        return user_pdf_name;
    }

    public void setUser_pdf_name(String user_pdf_name) {
        this.user_pdf_name = user_pdf_name;
    }

    public String getQuiz_content() {
        return quiz_content;
    }

    public void setQuiz_content(String quiz_content) {
        this.quiz_content = quiz_content;
    }

    public ChatEntity getDefChatEntity() {
        return defChatEntity;
    }

    public void setDefChatEntity(ChatEntity defChatEntity) {
        this.defChatEntity = defChatEntity;
    }


}
```

9.61 src/main/java/capstoneProject/Lorenzo/genIz/entity/OnlineModel↩ Entity.java File Reference

```
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
```

```
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
```

Classes

- class capstoneProject.Lorenzo.genIz.entity.OnlineModelEntity

Packages

- package capstoneProject.Lorenzo.genIz.entity

9.62 OnlineModelEntity.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.entity;
00002
00003 import jakarta.persistence.Column;
00004 import jakarta.persistence.Entity;
00005 import jakarta.persistence.GeneratedValue;
00006 import jakarta.persistence.GenerationType;
00007 import jakarta.persistence.Id;
00008 import jakarta.persistence.Table;
00009
00010 @Entity
00011 @Table(name = "online_model")
00012 public class OnlineModelEntity {
00013
00014     @Id
00015     @GeneratedValue(strategy=GenerationType.IDENTITY)
00016     @Column(name="model_id")
00017     private Integer id;
00018
00019     @Column(name="model_url")
00020     private String modelUrl;
00021
00022     @Column(name="model_name")
00023     private String modelName;
00024
00025     @Column(name="api_key")
00026     private String apiKey;
00027
00028     public Integer getId() {
00029         return id;
00030     }
00031
00032     public void setId(Integer id) {
00033         this.id = id;
00034     }
00035
00036     public String getModelUrl() {
00037         return modelUrl;
00038     }
00039
00040     public void setModelUrl(String modelUrl) {
00041         this.modelUrl = modelUrl;
00042     }
00043
00044     public String getModelName() {
00045         return modelName;
00046     }
00047
00048     public void setModelName(String modelName) {
00049         this.modelName = modelName;
00050     }
00051
00052     public String getApiKey() {
00053         return apiKey;
00054     }
00055
00056     public void setApiKey(String apiKey) {
00057         this.apiKey = apiKey;
00058     }
00059
00060 }
```

9.63 src/main/java/capstoneProject/Lorenzo/genIz/entity/UserEntity.java File Reference

```
import java.util.ArrayList;
import java.util.List;
import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.OneToMany;
import jakarta.persistence.Table;
```

Classes

- class capstoneProject.Lorenzo.genIz.entity.UserEntity

Packages

- package capstoneProject.Lorenzo.genIz.entity

9.64 UserEntity.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.entity;
00002
00003 import java.util.ArrayList;
00004 import java.util.List;
00005
00006 import jakarta.persistence.CascadeType;
00007 import jakarta.persistence.Column;
00008 import jakarta.persistence.Entity;
00009 import jakarta.persistence.GeneratedValue;
00010 import jakarta.persistence.GenerationType;
00011 import jakarta.persistence.Id;
00012 import jakarta.persistence.OneToMany;
00013 import jakarta.persistence.Table;
00014
00015 @Entity
00016 @Table(name="users")
00017 public class UserEntity {
00018
00019     @Id
00020     @GeneratedValue(strategy=GenerationType.IDENTITY)
00021     @Column(name = "id")
00022     private Integer id;
00023
00024     @Column(name="identity_provider_user_id")
00025     private String identity_provider_user_id;
00026
00027     @Column(name="use_local_model")
00028     private Boolean use_local_model;
00029
00030     //a user will have multiple chats (bidirectional connection)
00031     //this is the owned side
00032     @OneToMany(mappedBy="defUserEntity", cascade=CascadeType.ALL)
00033     private List<ChatEntity> chatEntityList;
00034
00035
00036     //method to add the ChatEntities
00037     public Integer getId() {
00038         return id;
00039     }
```

```java


    public void setId(Integer id) {
        this.id = id;
    }


    public String getIdentity_provider_user_id() {
        return identity_provider_user_id;
    }


    public void setIdentity_provider_user_id(String identity_provider_user_id) {
        this.identity_provider_user_id = identity_provider_user_id;
    }


    public List<ChatEntity> getChatEntityList() {
        return chatEntityList;
    }


    public void setChatEntityList(List<ChatEntity> chatEntityList) {
        this.chatEntityList = chatEntityList;
    }


    public Boolean getUse_local_model() {
        return use_local_model;
    }


    public void setUse_local_model(Boolean use_local_model) {
        this.use_local_model = use_local_model;
    }


    //method to add new ChatEntities to the list
    public void addChatEntity(ChatEntity chatEntity){
        if(chatEntityList == null){
            chatEntityList = new ArrayList<>();
        }

        chatEntityList.add(chatEntity);
    }
}
```

9.65 src/main/java/capstoneProject/Lorenzo/genIz/GenIzApplication.java File Reference

```java
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

Classes

- class capstoneProject.Lorenzo.genIz.GenIzApplication

Packages

- package capstoneProject.Lorenzo.genIz

9.66 GenIzApplication.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class GenIzApplication {

    public static void main(String[] args) {
        SpringApplication.run(GenIzApplication.class, args);
    }

}
```

9.67 src/main/java/capstoneProject/Lorenzo/genIz/global_exception_handler/GlobalExceptionHandler.java File Reference

```
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;
import org.springframework.web.method.annotation.MethodArgumentTypeMismatch↩
Exception;
import capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto;
import jakarta.persistence.NoResultException;
```

Classes

- class capstoneProject.Lorenzo.genIz.global_exception_handler.GlobalExceptionHandler

Packages

- package capstoneProject.Lorenzo.genIz.global_exception_handler

9.68 GlobalExceptionHandler.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.global_exception_handler;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;
import org.springframework.web.method.annotation.MethodArgumentTypeMismatchException;

import capstoneProject.Lorenzo.genIz.DTO.ErrorResponseDto;
import jakarta.persistence.NoResultException;

//global exception handler to manage the exceptions thrown by the code
@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(IllegalArgumentException.class)
    public ResponseEntity<ErrorResponseDto> handleIllegalArgumentException(IllegalArgumentException
    e){
```

```
        ErrorResponseDto error = new ErrorResponseDto(e.getMessage());
        return new ResponseEntity<>(error, HttpStatus.INTERNAL_SERVER_ERROR);
    }

    @ExceptionHandler(RuntimeException.class)
    public ResponseEntity<ErrorResponseDto> handleRuntimeException(RuntimeException e){
        ErrorResponseDto error = new ErrorResponseDto(e.getMessage());
        return new ResponseEntity<>(error, HttpStatus.INTERNAL_SERVER_ERROR);
    }

    @ExceptionHandler(NoResultException.class)
    public ResponseEntity<ErrorResponseDto> handleNoResultException(NoResultException e){
        ErrorResponseDto error = new ErrorResponseDto(e.getMessage());
        return new ResponseEntity<>(error, HttpStatus.INTERNAL_SERVER_ERROR);
    }

    @ExceptionHandler(MethodArgumentTypeMismatchException.class)
    public ResponseEntity<ErrorResponseDto> handleUnexpected(MethodArgumentTypeMismatchException ex) {
        ErrorResponseDto error = new ErrorResponseDto("The argument passed is not correct.");
        return new ResponseEntity<>(error, HttpStatus.INTERNAL_SERVER_ERROR);
    }

    // @ExceptionHandler(Exception.class)
    // public ResponseEntity<ErrorResponseDto> handleUnexpected(Exception ex) {
    //     ErrorResponseDto error = new ErrorResponseDto("An unexpected error occurred. Please try
    again later.");
    //     return new ResponseEntity<>(error, HttpStatus.INTERNAL_SERVER_ERROR);
    // }
}
```

9.69 src/main/java/capstoneProject/Lorenzo/genIz/security/Security↩ Config.java File Reference

```
import java.util.List;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.Customizer;
import org.springframework.security.config.annotation.web.builders.Http↩
Security;
import org.springframework.security.web.SecurityFilterChain;
import org.springframework.web.cors.CorsConfiguration;
import jakarta.servlet.http.HttpServletResponse;
```

Classes

- class capstoneProject.Lorenzo.genIz.security.SecurityConfig

Packages

- package capstoneProject.Lorenzo.genIz.security

9.70 SecurityConfig.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.security;

import java.util.List;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
```

```
import org.springframework.security.config.Customizer;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.web.SecurityFilterChain;
import org.springframework.web.cors.CorsConfiguration;

import jakarta.servlet.http.HttpServletResponse;

@Configuration
public class SecurityConfig {

    @Bean
    public SecurityFilterChain securityFilterChain(HttpSecurity http) throws Exception{
        return http.authorizeHttpRequests(
            securityConfig -> {
                securityConfig.requestMatchers("/").permitAll();
                securityConfig.anyRequest().authenticated();
            }
        )
        //return 401 error code if user is not authenticated
        .exceptionHandling(exception -> exception
            .authenticationEntryPoint((request, response, authException) -> {
                response.sendError(HttpServletResponse.SC_UNAUTHORIZED, "Unauthorized");
            }))

        //when user successfully loging, redirect to frontend home
        .oauth2Login(oauth2 -> oauth2.defaultSuccessUrl("http://localhost:24991/", true))

        .formLogin(Customizer.withDefaults())

        //logout config, by hitting baseUrl/logout the user gets redirected to a default logout page.
        .logout(logout -> {
            logout.logoutSuccessUrl("http://localhost:24991/")
            .permitAll()
            .logoutSuccessHandler((request, response, authentication) -> {
                response.setStatus(HttpServletResponse.SC_OK);
            });
        })

        .csrf(csrf -> csrf.disable())

        //allow home page of GenIz to perform api calls
        .cors(cors -> cors.configurationSource(request -> {
            CorsConfiguration config = new CorsConfiguration();
            config.setAllowedOrigins(List.of("http://localhost:24991/"));
            config.setAllowedMethods(List.of("GET", "POST", "PUT", "DELETE", "PATCH"));
            config.setAllowedHeaders(List.of("*"));
            config.setAllowCredentials(true);
            return config;
        }))

        .build();
    }
}
```

9.71 src/main/java/capstoneProject/Lorenzo/genIz/service/ApiFlow↩ AggregatorInterface.java File Reference

```
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto;
```

Classes

- interface capstoneProject.Lorenzo.genIz.service.ApiFlowAggregatorInterface

Packages

- package capstoneProject.Lorenzo.genIz.service

9.72 ApiFlowAggregatorInterface.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.service;

import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto;


public interface ApiFlowAggregatorInterface {

    //this service is used to merge the api call logic and the verification logic

    //so i need a method to merform the api call with the given text from the user
    public ResponseDataDto performApiCall(PostReqUserTextDto postReqUserTextDto);

    //and once the api request if prformet and response recieved i have to verify the result, and
    //if it's all good, i can return the needed dto to respond to ghe user.
    public ResponseDataDto validateApiCallResult(ResponseDataDto tempDTO);

}
```

9.73 src/main/java/capstoneProject/Lorenzo/genIz/service/ExportQuiz↩ ToLocalMachine.java File Reference

```
import java.util.List;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussion↩
ListDto;
```

Classes

- interface capstoneProject.Lorenzo.genIz.service.ExportQuizToLocalMachine

Packages

- package capstoneProject.Lorenzo.genIz.service

9.74 ExportQuizToLocalMachine.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.service;

import java.util.List;

import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;

public interface ExportQuizToLocalMachine {
    public List<String> formatFileContent(List<ResponseDiscussionListDto> quizList);

    public void saveFile(List<String> fileContent);
}
```

9.75 src/main/java/capstoneProject/Lorenzo/genIz/service/ExportTxt↩ File.java File Reference

```
import java.io.FileWriter;
import java.io.IOException;
import java.nio.file.Path;
import java.nio.file.Paths;
import java.time.LocalDateTime;
import java.util.ArrayList;
import java.util.List;
import org.springframework.stereotype.Service;
import com.google.gson.JsonArray;
import com.google.gson.JsonElement;
import com.google.gson.JsonObject;
import com.google.gson.JsonParser;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussion↩
ListDto;
```

Classes

- class capstoneProject.Lorenzo.genIz.service.ExportTxtFile

Packages

- package capstoneProject.Lorenzo.genIz.service

9.76 ExportTxtFile.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.service;
00002
00003 import java.io.FileWriter;
00004 import java.io.IOException;
00005 import java.nio.file.Path;
00006 import java.nio.file.Paths;
00007 import java.time.LocalDateTime;
00008 import java.util.ArrayList;
00009 import java.util.List;
00010
00011 import org.springframework.stereotype.Service;
00012
00013 import com.google.gson.JsonArray;
00014 import com.google.gson.JsonElement;
00015 import com.google.gson.JsonObject;
00016 import com.google.gson.JsonParser;
00017
00018 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;
00019
00020 @Service
00021 public class ExportTxtFile implements ExportQuizToLocalMachine{
00022
00023     @Override
00024     public List<String> formatFileContent(List<ResponseDiscussionListDto> quizList) {
00025
00026         List<String> discussionList = new ArrayList<>();
00027
00028         //create the string object that contains the question data
00029         for(ResponseDiscussionListDto responseDiscussionListDto : quizList){
00030
00031             //get hold of the quiz content
00032             String json = responseDiscussionListDto.getQuiz_content();
00033
```

```
            //extract the json object
            JsonObject rootJson = JsonParser.parseString(json).getAsJsonObject();

            //extract the json list
            JsonArray quizzes = rootJson.getAsJsonArray("quizzes");

            //loop inside the list and append the element to the Stringbuiler element
            //then add the element to the discussionList
            for(JsonElement jsonElement : quizzes){

                //create the stringBuilder element
                StringBuilder quizBuilder = new StringBuilder();

                JsonObject quizElements = jsonElement.getAsJsonObject();
                JsonObject options = quizElements.getAsJsonObject("options");

                quizBuilder.append(quizElements.get("question")).append(System.lineSeparator());
                quizBuilder.append("A: ").append(options.get("A")).append(System.lineSeparator());
                quizBuilder.append("B: ").append(options.get("B")).append(System.lineSeparator());
                quizBuilder.append("C: ").append(options.get("C")).append(System.lineSeparator());
                quizBuilder.append("D: ").append(options.get("D")).append(System.lineSeparator());
                quizBuilder.append("Correct answer: ").append(quizElements.get("correct_answer")).append(System.lineSeparator());
                quizBuilder.append("Explanation: ").append(quizElements.get("explanation")).append(System.lineSeparator());
                quizBuilder.append(System.lineSeparator());
                quizBuilder.append(System.lineSeparator());

                discussionList.add(quizBuilder.toString());
            }
        }

        return discussionList;
    }

    @Override
    public void saveFile(List<String> fileContent){

        String fileName = LocalDateTime.now().toString().replace(":", "-") + ".txt";

        Path path = Paths.get("/exported_quizzes").resolve(fileName);

        //take each formatted quiz (if more than 1) and append it at the end of the file
        try (FileWriter writer = new FileWriter(path.toFile(), true)){
            for (String quiz : fileContent){
                writer.write(quiz);
            }
        } catch (IOException e){
            throw new RuntimeException("The txt file creation failed, please try again.");
        }
    }

}
```

9.77 src/main/java/capstoneProject/Lorenzo/genIz/service/GenerateQuizApi.java File Reference

```
import java.io.IOException;
import java.net.URI;
import java.net.URISyntaxException;
import java.net.http.HttpClient;
import java.net.http.HttpRequest;
import java.net.http.HttpResponse;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Service;
import com.google.gson.Gson;
import com.google.gson.JsonSyntaxException;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;
import capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest;
import capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse;
import jakarta.annotation.PostConstruct;
```

Classes

- class capstoneProject.Lorenzo.genIz.service.GenerateQuizApi

Packages

- package capstoneProject.Lorenzo.genIz.service

9.78 GenerateQuizApi.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.service;
00002
00003 import java.io.IOException;
00004 import java.net.URI;
00005 import java.net.URISyntaxException;
00006 import java.net.http.HttpClient;
00007 import java.net.http.HttpRequest;
00008 import java.net.http.HttpResponse;
00009
00010 import org.springframework.beans.factory.annotation.Value;
00011 import org.springframework.stereotype.Service;
00012
00013 import com.google.gson.Gson;
00014 import com.google.gson.JsonSyntaxException;
00015
00016 import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
00017 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;
00018 import capstoneProject.Lorenzo.genIz.api_format.request.ApiCallRequest;
00019 import capstoneProject.Lorenzo.genIz.api_format.response.ApiCallResponse;
00020 import jakarta.annotation.PostConstruct;
00021
00022 @Service
00023 public class GenerateQuizApi implements GenerateQuizApiInterface{
00024
00025     //injcet database service to obtain online model data
00026     QuizDaoService quizDaoService;
00027
00028     public GenerateQuizApi(QuizDaoService quizDaoService){
00029         this.quizDaoService = quizDaoService;
00030     }
00031
00032     //injecting info from .env file
00033     @Value("${LLM_MODEL}")
00034     private String modelName;
00035
00036     @Value("${MODEL_URL}")
00037     private String modelUrl;
00038
00039     private ResponseOnlineModelDto responseOnlineModelDto;
00040
00041     //populate the DTO after class initialization (neede because quizDaoService must be created before
      this query)
00042     @PostConstruct
00043     private void retrieveOnlineModelSettings(){
00044         try {
00045             responseOnlineModelDto = quizDaoService.retrieveOnlineModelSettings();
00046         } catch (Exception e) {
00047             responseOnlineModelDto = new ResponseOnlineModelDto();
00048             responseOnlineModelDto.setModel_url("null");
00049             responseOnlineModelDto.setModel_name("null");
00050             responseOnlineModelDto.setApi_key("null");
00051         }
00052     }
00053
00054     //initializing the HTTP client
00055     HttpClient httpClient = HttpClient.newHttpClient();
00056
00057     //initializing Gson
00058     Gson gson = new Gson();
00059
00060
00061     //creating the API request json body
00062     @Override
00063     public String createApiBody(String systemPrompt, String userText, Boolean useLocalModel) {
00064
```

```
        //creating the api body
        ApiCallRequest apiCallRequest = new ApiCallRequest();
        if(!useLocalModel){
            apiCallRequest.setModel(responseOnlineModelDto.getModel_name());
        } else {
            apiCallRequest.setModel(modelName);
        }
        apiCallRequest.addField("system", systemPrompt);
        apiCallRequest.addField("user", userText);

        //map the class to it's json string
        String json = gson.toJson(apiCallRequest);

        return json;
    }

    //creating the HTTP request to query the model
    @Override
    public HttpRequest apiCallRequest(String requestBody, Boolean useLocalModel){

        //if useLocal model is not set, use local model
        if(useLocalModel == null){
            useLocalModel = true;
        }

        //initializing the request
        HttpRequest httpRequest = null;

        //if user set useLocalModel to false, use the specified model provier to create the request, else use local model
        if(!useLocalModel){
            try {
                httpRequest = HttpRequest.newBuilder()
                    .uri(new URI(responseOnlineModelDto.getModel_url()))
                    .header("Content-Type", "application/json")
                    .header("Authorization", "Bearer " + responseOnlineModelDto.getApi_key())
                    .POST(HttpRequest.BodyPublishers.ofString(requestBody))
                    .build();

            } catch (URISyntaxException e) {
                throw new RuntimeException("Something went wrong with the model url", e);
            }

        } else {
            try {
                httpRequest = HttpRequest.newBuilder()
                    .uri(new URI(modelUrl))
                    .header("Content-Type", "application/json")
                    .POST(HttpRequest.BodyPublishers.ofString(requestBody))
                    .build();

            } catch (URISyntaxException e) {
                throw new RuntimeException("Something went wrong with the model url", e);
            }
        }

        return httpRequest;
    }

    //retrieving the response by sending the request to the model
    @Override
    public ResponseDataDto getApiResponse(HttpRequest apiRequest){
        //using the http client to send the request to the running model
        HttpResponse<String> rawResponse;
        try{
            rawResponse = httpClient.send(apiRequest, HttpResponse.BodyHandlers.ofString());

            //converting the response to a DTO object
            ApiCallResponse convertResponse = gson.fromJson(rawResponse.body(), ApiCallResponse.class);

            return new ResponseDataDto(convertResponse.getChoices().get(0).getMessage().getContent(),
             convertResponse.getUsage().getTotal_tokens());

        } catch (IOException | InterruptedException | JsonSyntaxException e){
            throw new RuntimeException("Something went wrong when calling the llm model or with the model response format", e);
        }
    }

}
```

9.79 src/main/java/capstoneProject/Lorenzo/genIz/service/GenerateQuizApiInterface.java File Reference

```
import java.net.http.HttpRequest;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
```

Classes

- interface capstoneProject.Lorenzo.genIz.service.GenerateQuizApiInterface

Packages

- package capstoneProject.Lorenzo.genIz.service

9.80 GenerateQuizApiInterface.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.service;
00002
00003 import java.net.http.HttpRequest;
00004
00005 import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
00006
00007 public interface GenerateQuizApiInterface {
00008
00009     public String createApiBody(String systemPrompt, String userText, Boolean useLocalModel);
00010
00011     public HttpRequest apiCallRequest(String requestBody, Boolean useLocalModel);
00012
00013     public ResponseDataDto getApiResponse(HttpRequest apiRequest);
00014 }
```

9.81 src/main/java/capstoneProject/Lorenzo/genIz/service/ImplementQuizDaoMethodsInterface.java File Reference

```
import java.util.List;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussion↩
ListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSetting↩
Dto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModel↩
Dto;
```

Classes

- interface capstoneProject.Lorenzo.genIz.service.ImplementQuizDaoMethodsInterface

Packages

- package capstoneProject.Lorenzo.genIz.service

9.82 ImplementQuizDaoMethodsInterface.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.service;


import java.util.List;

import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;

public interface ImplementQuizDaoMethodsInterface {

    public ResponseChatListDto saveChat(PostReqChatDto postReqChatDto);

    public List<ResponseChatListDto> retrieveChats();

    public ResponseDiscussionListDto saveDiscussion(PostReqDiscussionDto postReqDiscussionDto);

    public List<ResponseDiscussionListDto> retrieveDiscussions(Integer chatId);

    public ResponseModelSettingDto manageModelSettings(PostReqModelSetting postReqModelSetting);

    public ResponseModelSettingDto retrieveModelSettings();

    public void deleteChat(Integer chatId);

    public ResponseChatListDto updateChat(PostReqChatDto postReqChatDto);

    public void deleteDiscussion(Integer discussionId);

    public void saveNewOnlineModelSettings(PostReqOnlineModelDto postReqOnlineModelDto);

    public ResponseOnlineModelDto retrieveOnlineModelSettings();

    public void deleteOnlineModelSettings();
}
```

9.83 src/main/java/capstoneProject/Lorenzo/genIz/service/Json↩ Validation.java File Reference

```
import org.springframework.stereotype.Service;
import com.google.gson.JsonArray;
import com.google.gson.JsonElement;
import com.google.gson.JsonObject;
import com.google.gson.JsonParser;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
```

Classes

- class capstoneProject.Lorenzo.genIz.service.JsonValidation

Packages

- package capstoneProject.Lorenzo.genIz.service

9.84 JsonValidation.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.service;

import org.springframework.stereotype.Service;

import com.google.gson.JsonArray;
import com.google.gson.JsonElement;
import com.google.gson.JsonObject;
import com.google.gson.JsonParser;

import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;

@Service
public class JsonValidation implements JsonValidationInterface{

    //retrieving the DTO created by the API call and transform it into a json object
    @Override
    public JsonObject convertStringToJsonObject(ResponseDataDto quizDataDto) {

        //json returned by the model
        //since tokenCount is not a critical information, it's accettable to let the application work even without it
        String jsonLlmOutput = quizDataDto.getLlmOutput();

        //if the json is incorrect, parseString will return an exception JsonSyntaxException
        JsonObject rootJson = JsonParser.parseString(jsonLlmOutput).getAsJsonObject();

        return rootJson;
    }

    //validating the fields inside the API response
    @Override
    public ResponseDataDto validateJsonObject(JsonObject objectToValidate, ResponseDataDto quizDataDto) {

        //retrieving the quizzes array and validate content
        JsonArray quizList = objectToValidate.getAsJsonArray("quizzes");

        for (JsonElement x : quizList){
            JsonObject quizBody = x.getAsJsonObject();

            //validate each quiz present inside the response
            if (!quizBody.has("question")){throw new IllegalArgumentException("question field has problems");}
            if (!quizBody.has("options")){throw new IllegalArgumentException("options fiels has problems");}

            JsonObject questionOptions = quizBody.getAsJsonObject("options");
            if (!questionOptions.has("A")){throw new IllegalArgumentException("a question option has a problem");}
            if (!questionOptions.has("B")){throw new IllegalArgumentException("a question option has a problem");}
            if (!questionOptions.has("C")){throw new IllegalArgumentException("a question option has a problem");}
            if (!questionOptions.has("D")){throw new IllegalArgumentException("a question option has a problem");}

            if (!quizBody.has("explanation")){throw new IllegalArgumentException("explanation has a problem");}
            if (!quizBody.has("correct_answer")){throw new IllegalArgumentException("answer has a problem");}
        }

        //if all checks are passed, return the verified DTO
        return quizDataDto;
    }

}
```

9.85 src/main/java/capstoneProject/Lorenzo/genIz/service/JsonValidationInterface.java File Reference

```
import com.google.gson.JsonObject;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
```

Classes

- interface capstoneProject.Lorenzo.genIz.service.JsonValidationInterface

Packages

- package capstoneProject.Lorenzo.genIz.service

9.86 JsonValidationInterface.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.service;
00002
00003 import com.google.gson.JsonObject;
00004
00005 import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
00006
00007 public interface JsonValidationInterface {
00008
00009     public JsonObject convertStringToJsonObject(ResponseDataDto quizDataDto);
00010
00011     public ResponseDataDto validateJsonObject(JsonObject objectToValidate, ResponseDataDto
      quizDataDto);
00012
00013 }
```

9.87 src/main/java/capstoneProject/Lorenzo/genIz/service/LocalModelApiFlowAggregator.java File Reference

```
import java.net.http.HttpRequest;
import org.springframework.stereotype.Service;
import com.google.gson.JsonObject;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto;
import capstoneProject.Lorenzo.genIz.api_format.request.request_parameters.↩
QuizGenParameter;
```

Classes

- class capstoneProject.Lorenzo.genIz.service.LocalModelApiFlowAggregator

Packages

- package capstoneProject.Lorenzo.genIz.service

9.88 LocalModelApiFlowAggregator.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz.service;

import java.net.http.HttpRequest;

import org.springframework.stereotype.Service;

import com.google.gson.JsonObject;

import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqUserTextDto;
import capstoneProject.Lorenzo.genIz.api_format.request.request_parameters.QuizGenParameter;

@Service
public class LocalModelApiFlowAggregator implements ApiFlowAggregatorInterface{

    //injecting services needed to perform business logic
    GenerateQuizApi generateQuizApi;
    JsonValidation jsonValidation;

    public LocalModelApiFlowAggregator(GenerateQuizApi generateQuizApi, JsonValidation jsonValidation) {
        this.generateQuizApi = generateQuizApi;
        this.jsonValidation = jsonValidation;
    }

    @Override
    public ResponseDataDto performApiCall(PostReqUserTextDto postReqUserTextDto) {

        //creating the api body
        String apiBody = generateQuizApi.createApiBody(QuizGenParameter.getSystemprompt(), postReqUserTextDto.getUserText(), postReqUserTextDto.getUseLocalModel());

        //creating the api request
        HttpRequest apiRequest = generateQuizApi.apiCallRequest(apiBody, postReqUserTextDto.getUseLocalModel());

        //send the api request and get a response
        ResponseDataDto apiResponse = generateQuizApi.getApiResponse(apiRequest);

        return apiResponse;
    }

    @Override
    public ResponseDataDto validateApiCallResult(ResponseDataDto tempDTO) {

        //convert the llmOutput from the dto into json format for parsing
        JsonObject llmOutputConvertedToJson = jsonValidation.convertStringToJsonObject(tempDTO);

        //validate the json format to ensure data presence
        ResponseDataDto validatedModelResponse = jsonValidation.validateJsonObject(llmOutputConvertedToJson, tempDTO);

        return validatedModelResponse;
    }

}
```

9.89 src/main/java/capstoneProject/Lorenzo/genIz/service/QuizDao↩ Service.java File Reference

```
import java.util.ArrayList;
import java.util.List;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.oauth2.client.authentication.OAuth2↩
AuthenticationToken;
import org.springframework.stereotype.Service;
import capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto;
import capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO;
```

```
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussion↩
ListDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSetting↩
Dto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModel↩
Dto;
```

Classes

- class capstoneProject.Lorenzo.genIz.service.QuizDaoService

Packages

- package capstoneProject.Lorenzo.genIz.service

9.90 QuizDaoService.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz.service;
00002
00003 import java.util.ArrayList;
00004 import java.util.List;
00005
00006 import org.springframework.security.core.context.SecurityContextHolder;
00007 import org.springframework.security.oauth2.client.authentication.OAuth2AuthenticationToken;
00008 import org.springframework.stereotype.Service;
00009
00010 import capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl;
00011 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.ChatDataDto;
00012 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.DiscussionDataDto;
00013 import capstoneProject.Lorenzo.genIz.DTO.entity_dto.UserDataDTO;
00014 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqChatDto;
00015 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqDiscussionDto;
00016 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqModelSetting;
00017 import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
00018 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseChatListDto;
00019 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseDiscussionListDto;
00020 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseModelSettingDto;
00021 import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;
00022
00023 @Service
00024 public class QuizDaoService implements ImplementQuizDaoMethodsInterface{
00025
00026     //injecting the quizDao implementation
00027     QuizDaoImpl quizDaoImpl;
00028
00029     public QuizDaoService(QuizDaoImpl quizDaoImpl) {
00030         this.quizDaoImpl = quizDaoImpl;
00031     }
00032
00033     //creating the function to retrieve the Oauth2 token when needed
00034     public OAuth2AuthenticationToken retrieveToken(){
00035         return (OAuth2AuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
00036     }
00037
00038     @Override
00039     public ResponseChatListDto saveChat(PostReqChatDto postReqChatDto) {
00040
00041         UserDataDTO currentUser = quizDaoImpl.manageUser(retrieveToken());
00042
00043         //check if chat_name is defined in postReqChatDto, if not stop the execution
00044         if(postReqChatDto.getChatName() == null){
```

```
            throw new IllegalArgumentException("chat name must be defined");
        }

        ChatDataDto currentChat = quizDaoImpl.saveChat(currentUser, postReqChatDto);

        //map the returned chat dto to a response dto
        ResponseChatListDto responseChatDto = new ResponseChatListDto();
        responseChatDto.setChat_id(currentChat.getChat_id());
        responseChatDto.setChat_name(currentChat.getChat_name());

        return responseChatDto;
    }

    @Override
    public List<ResponseChatListDto> retrieveChats() {

        UserDataDTO currentUser = quizDaoImpl.manageUser(retrieveToken());

        //check if the user id is present in userDataDTO, if not stop the execution
        if(currentUser.getId() == null){
            throw new IllegalArgumentException("user id must be defined");
        }

        List<ChatDataDto> currentChats = quizDaoImpl.retrieveChats(currentUser);

        //map the list of chat dto to the response dto that will be sent to the user
        List<ResponseChatListDto> responseChatsDto = new ArrayList<>();
        for(ChatDataDto chat : currentChats){
            ResponseChatListDto responseChat = new ResponseChatListDto();
            responseChat.setChat_id(chat.getChat_id());
            responseChat.setChat_name(chat.getChat_name());
            responseChatsDto.add(responseChat);
        }

        return responseChatsDto;
    }

    @Override
    public ResponseDiscussionListDto saveDiscussion(PostReqDiscussionDto postReqDiscussionDto) {

        //checking the postReqDiscussionDto fields, if not present stop the execution
        if(postReqDiscussionDto.getChat_id()==null){throw new IllegalArgumentException("chat id must be defined");}
        if(postReqDiscussionDto.getUser_pdf_name()==null){throw new IllegalArgumentException("pdf name must be defined");}
        if(postReqDiscussionDto.getQuiz_content()==null){throw new IllegalArgumentException("quiz content must be defined");}

        DiscussionDataDto currentDuscussion = quizDaoImpl.saveDiscussion(postReqDiscussionDto);

        //map the returned discussion dto into a response dto
        ResponseDiscussionListDto responseDiscussiondto = new ResponseDiscussionListDto();
        responseDiscussiondto.setDiscussion_id(currentDuscussion.getDiscussion_id());
        responseDiscussiondto.setUser_pdf_name(currentDuscussion.getUser_pdf_name());
        responseDiscussiondto.setQuiz_content(currentDuscussion.getQuiz_content());

        return responseDiscussiondto;
    }

    @Override
    public List<ResponseDiscussionListDto> retrieveDiscussions(Integer chatId) {

        //check if the chat id is defined, if not stop execution
        if(chatId == null){
            throw new IllegalArgumentException("chat id must be defined");
        }

        List<DiscussionDataDto> currentDiscussions = quizDaoImpl.retrieveDiscussions(chatId);

        //map the returned list of discussion dtos into a list of response dto
        List<ResponseDiscussionListDto> responseDiscussionsDto = new ArrayList<>();
        for(DiscussionDataDto discussion : currentDiscussions){
            ResponseDiscussionListDto responseDiscussion = new ResponseDiscussionListDto();
            responseDiscussion.setDiscussion_id(discussion.getDiscussion_id());
            responseDiscussion.setUser_pdf_name(discussion.getUser_pdf_name());
            responseDiscussion.setQuiz_content(discussion.getQuiz_content());
            responseDiscussionsDto.add(responseDiscussion);
        }

        return responseDiscussionsDto;
    }

    @Override
    public ResponseModelSettingDto manageModelSettings(PostReqModelSetting postReqModelSetting) {

        UserDataDTO currentUser = quizDaoImpl.manageUser(retrieveToken());

```

```
        //check if user id and model setting have been defined, if not stop execution
        if(currentUser.getId()==null){throw new IllegalArgumentException("user id must be defined");}
        if(postReqModelSetting.getModelSetting()==null){throw new IllegalArgumentException("model setting must be defined");}

        UserDataDTO newCurrentUser = quizDaoImpl.manageModelSettings(currentUser, postReqModelSetting);

        //map the current user to return the model setting repsonse
        ResponseModelSettingDto responseModelSettingDto = new ResponseModelSettingDto();
        responseModelSettingDto.setUse_local_model(newCurrentUser.getUse_local_model());

        return responseModelSettingDto;
    }

    @Override
    public ResponseModelSettingDto retrieveModelSettings() {

        UserDataDTO currentUser = quizDaoImpl.manageUser(retrieveToken());

        //check if the user id is defined, if not stop execution
        if(currentUser.getId() == null){
            throw new IllegalArgumentException("user id must be defined");
        }

        UserDataDTO newCurrentUser = quizDaoImpl.retrieveModelSettings(currentUser);

        //map the current user to return the model setting repsonse
        ResponseModelSettingDto responseModelSettingDto = new ResponseModelSettingDto();
        responseModelSettingDto.setUse_local_model(newCurrentUser.getUse_local_model());

        return responseModelSettingDto;
    }

    @Override
    public void deleteChat(Integer chatId) {

        //check if chatId id defined, if not stop execution
        if(chatId == null){
            throw new IllegalArgumentException("chat id must be defined");
        }

        //call the delete method
        quizDaoImpl.deleteChat(chatId);
    }

    @Override
    public ResponseChatListDto updateChat(PostReqChatDto postReqChatDto) {

        //check if the chat id and chat name are defined
        if(postReqChatDto.getChat_id()==null){throw new IllegalArgumentException("chat id must be defined");}
        if(postReqChatDto.getChatName()==null){throw new IllegalArgumentException("chat name must be defined");}

        //update the chat by using the dao method
        ChatDataDto updatedChat = quizDaoImpl.updateChat(postReqChatDto);

        //map the updated chat to the resposne dto
        ResponseChatListDto responseChatDto = new ResponseChatListDto();
        responseChatDto.setChat_id(updatedChat.getChat_id());
        responseChatDto.setChat_name(updatedChat.getChat_name());

        return responseChatDto;
    }

    @Override
    public void deleteDiscussion(Integer discussionId) {

        //check if discussion id is defined, if not stop execution
        if(discussionId == null){
            throw new IllegalArgumentException("discussion id must be defined");
        }

        //call the delete method
        quizDaoImpl.deleteDiscussion(discussionId);
    }

    @Override
    public void saveNewOnlineModelSettings(PostReqOnlineModelDto postReqOnlineModelDto) {

        //no strict format requirements
        quizDaoImpl.saveNewOnlineModelSettings(postReqOnlineModelDto);
    }

    @Override
    public ResponseOnlineModelDto retrieveOnlineModelSettings() {
```

```
        ResponseOnlineModelDto responseOnlineModelDto = quizDaoImpl.retrieveOnlineModelSettings();

        //validate data existance
        if(responseOnlineModelDto.getModel_url() == null || responseOnlineModelDto.getModel_url() ==
    ""){throw new IllegalArgumentException("Model Url is not defined.");}
        if(responseOnlineModelDto.getModel_name() == null || responseOnlineModelDto.getModel_name() ==
    ""){throw new IllegalArgumentException("Model Name is not defined.");}
        if(responseOnlineModelDto.getApi_key() == null || responseOnlineModelDto.getApi_key() ==
    ""){throw new IllegalArgumentException("Api Key not defined.");}

        return responseOnlineModelDto;
    }

    @Override
    public void deleteOnlineModelSettings() {

        //no strict format requirements
        quizDaoImpl.deleteOnlineModelSettings();
    }

}
```

9.91 src/test/java/capstoneProject/Lorenzo/genIz/GenIzApplication↩ Tests.java File Reference

```
import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
```

Classes

- class capstoneProject.Lorenzo.genIz.GenIzApplicationTests

Packages

- package capstoneProject.Lorenzo.genIz

9.92 GenIzApplicationTests.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
class GenIzApplicationTests {

    @Test
    void contextLoads() {
    }

}
```

9.93 src/test/java/capstoneProject/Lorenzo/genIz/JsonValidation↩ Test.java File Reference

```
import org.junit.jupiter.api.Assertions.assertDoesNotThrow;
import org.junit.jupiter.api.Assertions.assertThrows;
import org.junit.jupiter.api.Test;
import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
import capstoneProject.Lorenzo.genIz.service.JsonValidation;
```

Classes

- class capstoneProject.Lorenzo.genIz.JsonValidationTest

Packages

- package capstoneProject.Lorenzo.genIz

9.94 JsonValidationTest.java

Go to the documentation of this file.

```
00001 package capstoneProject.Lorenzo.genIz;
00002
00003 import static org.junit.jupiter.api.Assertions.assertDoesNotThrow;
00004 import static org.junit.jupiter.api.Assertions.assertThrows;
00005 import org.junit.jupiter.api.Test;
00006
00007 import capstoneProject.Lorenzo.genIz.DTO.ResponseDataDto;
00008 import capstoneProject.Lorenzo.genIz.service.JsonValidation;
00009
00010 public class JsonValidationTest {
00011
00012     //run test with correct json to verify code functionality
00013     @Test
00014     void testCorrectExecutionOfTheValidationSteps(){
00015         String goodApiResponse = "{\"quizzes\":[{\"question\":\"In a Directed Acyclic Graph (DAG),
      which of the following best describes the flow of tasks?\",\"options\":{\"A\":\"Tasks can only proceed
      in one direction, and there are no loops or cycles.\",\"B\":\"Tasks can move in any direction,
      including backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles
      are allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to
      ensure determinism.\"},\"explanation\":\"In a DAG, the flow is unidirectional, meaning tasks can only
      proceed in one direction (from the start to the end). There are no loops or cycles allowed, which
      ensures that the workflow can reach a completed
      state.\",\"correct_answer\":\"D\"},{\"question\":\"What does the term 'acyclic' in Directed Acyclic
      Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never
      backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no
      loops.\",\"C\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are
      executed in a single direction, but cycles are allowed for better resource
      utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a
      task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and
      avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct
      example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform
      \\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract
      from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C
      Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks
      to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run
      in parallel before loading to Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A
      depends on Task B, what happens if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because
      of the dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect
      it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate
      downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is
      completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still
      continue to run as Task A's failure does not affect it. However, the failure will propagate
      downstream, potentially causing issues with downstream tasks if they depend on Task A's
      results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a
      fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C
      Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C
```

```
Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C
Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C
Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before
a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform)
are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How
does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks
can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction,
including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By
ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs
prevents loops, meaning tasks can only move in one direction (from start to end). This prevents
infinite loops where a task might keep repeating a previous state due to
cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertDoesNotThrow( () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto),
responseDataDto);
            }
        );
    }

    //test to validate the quizzes array
    @Test
    void testValidationStepsWithMissingQuizzesArray(){
        String goodApiResponse = "{\"quiz\":[{\"question\":\"In a Directed Acyclic Graph (DAG), which
of the following best describes the flow of tasks?\",\"options\":{\"A\":\"Tasks can only proceed in
one direction, and there are no loops or cycles.\",\"B\":\"Tasks can move in any direction, including
backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles are
allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to
ensure determinism.\"},\"explanation\":\"In a DAG, the flow is unidirectional, meaning tasks can only
proceed in one direction (from the start to the end). There are no loops or cycles allowed, which
ensures that the workflow can reach a completed
state.\",\"correct_answer\":\"D\"},{\"question\":\"What does the term 'acyclic' in Directed Acyclic
Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never
backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no
loops.\",\"C\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are
executed in a single direction, but cycles are allowed for better resource
utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a
task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and
avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct
example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform
\\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract
from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C
Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks
to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run
in parallel before loading to Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A
depends on Task B, what happens if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because
of the dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect
it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate
downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is
completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still
continue to run as Task A's failure does not affect it. However, the failure will propagate
downstream, potentially causing issues with downstream tasks if they depend on Task A's
results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a
fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C
Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C
Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C
Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C
Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before
a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform)
are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How
does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks
can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction,
including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By
ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs
prevents loops, meaning tasks can only move in one direction (from start to end). This prevents
infinite loops where a task might keep repeating a previous state due to
cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertThrows(NullPointerException.class,
                () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto),
responseDataDto);
            }
        );
    }

    //test to validate the question section
    @Test
    void testValidationStepsWithMissingOrMispelledQuestionSection(){
```

```
        String goodApiResponse = "{\"quizzes\":[{\"\":\"In a Directed Acyclic Graph (DAG), which of the following best describes the flow of tasks?\",\"options\":{\"A\":\"Tasks can only proceed in one direction, and there are no loops or cycles.\",\"B\":\"Tasks can move in any direction, including backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles are allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to ensure determinism.\"},\"explanation\":\"In a DAG, the flow is unidirectional, meaning tasks can only proceed in one direction (from the start to the end). There are no loops or cycles allowed, which ensures that the workflow can reach a completed state.\",\"correct_answer\":\"D\"},{\"questions\":\"What does the term 'acyclic' in Directed Acyclic Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no loops.\",\"C\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are executed in a single direction, but cycles are allowed for better resource utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run in parallel before loading to Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A depends on Task B, what happens if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because of the dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still continue to run as Task A's failure does not affect it. However, the failure will propagate downstream, potentially causing issues with downstream tasks if they depend on Task A's results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform) are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction, including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs prevents loops, meaning tasks can only move in one direction (from start to end). This prevents infinite loops where a task might keep repeating a previous state due to cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertThrows(IllegalArgumentException.class,
            () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto), responseDataDto);
            }
        );
    }

    //test to validate the options section
    @Test
    void testValidationStepsWithMissingOrMispelledOptionsSection(){
        String goodApiResponse = "{\"quizzes\":[{\"question\":\"In a Directed Acyclic Graph (DAG), which of the following best describes the flow of tasks?\",\"responses\":{\"A\":\"Tasks can only proceed in one direction, and there are no loops or cycles.\",\"B\":\"Tasks can move in any direction, including backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles are allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to ensure determinism.\"},\"explanation\":\"In a DAG, the flow is unidirectional, meaning tasks can only proceed in one direction (from the start to the end). There are no loops or cycles allowed, which ensures that the workflow can reach a completed state.\",\"correct_answer\":\"D\"},{\"question\":\"What does the term 'acyclic' in Directed Acyclic Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no loops.\",\"C\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are executed in a single direction, but cycles are allowed for better resource utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run in parallel before loading to Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A depends on Task B, what happens if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because of the dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still
```

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continue to run as Task A's failure does not affect it. However, the failure will propagate
downstream, potentially causing issues with downstream tasks if they depend on Task A's
results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a
fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C
Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C
Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C
Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C
Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before
a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform)
are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How
does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks
can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction,
including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By
ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs
prevents loops, meaning tasks can only move in one direction (from start to end). This prevents
infinite loops where a task might keep repeating a previous state due to
cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertThrows(IllegalArgumentException.class,
            () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto),
responseDataDto);
            }
        );
    }

    //test to validate responses section
    @Test
    void testValidationStepsWithMissingOrMispelledOptionLetter(){
        String goodApiResponse = "{\"quizzes\":[{\"question\":\"In a Directed Acyclic Graph (DAG),
which of the following best describes the flow of tasks?\",\"options\":{\"a\":\"Tasks can only proceed
in one direction, and there are no loops or cycles.\",\"XX\":\"Tasks can move in any direction,
including backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles
are allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to
ensure determinism.\"},\"explanation\":\"In a DAG, the flow is unidirectional, meaning tasks can only
proceed in one direction (from the start to the end). There are no loops or cycles allowed, which
ensures that the workflow can reach a completed
state.\",\"correct_answer\":\"D\"},{\"question\":\"What does the term 'acyclic' in Directed Acyclic
Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never
backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no
loops.\",\"C2\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are
executed in a single direction, but cycles are allowed for better resource
utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a
task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and
avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct
example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform
\\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract
from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C
Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks
to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run
in parallel before loading to Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A
depends on Task B, what happens if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because
of the dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect
it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate
downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is
completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still
continue to run as Task A's failure does not affect it. However, the failure will propagate
downstream, potentially causing issues with downstream tasks if they depend on Task A's
results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a
fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C
Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C
Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C
Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C
Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before
a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform)
are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How
does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks
can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction,
including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By
ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs
prevents loops, meaning tasks can only move in one direction (from start to end). This prevents
infinite loops where a task might keep repeating a previous state due to
cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertThrows(IllegalArgumentException.class,
            () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto),
responseDataDto);
            }
```

```
        );
    }

    //test to validate explanation section
    @Test
    void testValidationStepsWithMissingOrMispelledExplanationSection(){
        String goodApiResponse = "{\"quizzes\":[{\"question\":\"In a Directed Acyclic Graph (DAG), which of the following best describes the flow of tasks?\",\"options\":{\"A\":\"Tasks can only proceed in one direction, and there are no loops or cycles.\",\"B\":\"Tasks can move in any direction, including backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles are allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to ensure determinism.\"},\"why_its_correct\":\"In a DAG, the flow is unidirectional, meaning tasks can only proceed in one direction (from the start to the end). There are no loops or cycles allowed, which ensures that the workflow can reach a completed state.\",\"correct_answer\":\"D\"},{\"question\":\"What does the term 'acyclic' in Directed Acyclic Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no loops.\",\"C\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are executed in a single direction, but cycles are allowed for better resource utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run in parallel before loading to Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A depends on Task B, what happens if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because of the dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still continue to run as Task A's failure does not affect it. However, the failure will propagate downstream, potentially causing issues with downstream tasks if they depend on Task A's results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform) are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction, including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs prevents loops, meaning tasks can only move in one direction (from start to end). This prevents infinite loops where a task might keep repeating a previous state due to cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertThrows(IllegalArgumentException.class,
            () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto), responseDataDto);
            }
        );
    }

    //test to validate answer section
    @Test
    void testValidationStepsWithMissingOrMispelledAnswerSection(){
        String goodApiResponse = "{\"quizzes\":[{\"question\":\"In a Directed Acyclic Graph (DAG), which of the following best describes the flow of tasks?\",\"options\":{\"A\":\"Tasks can only proceed in one direction, and there are no loops or cycles.\",\"B\":\"Tasks can move in any direction, including backward, and can have loops.\",\"C\":\"Tasks are executed in a single direction, but cycles are allowed for optimization.\",\"D\":\"Tasks flow in a single direction, but cycles are prevented to ensure determinism.\"},\"explanation\":\"In a DAG, the flow is unidirectional, meaning tasks can only proceed in one direction (from the start to the end). There are no loops or cycles allowed, which ensures that the workflow can reach a completed state.\",\"answer\":\"D\"},{\"question\":\"What does the term 'acyclic' in Directed Acyclic Graph (DAG) mean?\",\"options\":{\"A\":\"Tasks can only go in one direction, never backward.\",\"B\":\"Tasks can move in any direction including backward, and there are no loops.\",\"C\":\"Tasks have cycles or loops that allow for concurrent processing.\",\"D\":\"Tasks are executed in a single direction, but cycles are allowed for better resource utilization.\"},\"explanation\":\"Acyclic means that there are no loops or cycles in the graph. Once a task is done, it cannot start again, ensuring that the workflow can resolve to a completed state and avoid deadlocks.\",\"correct_answer\":\"D\"},{\"question\":\"Which of the following is a correct example of a fan-out topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform\",\"D\":\"Extract from S3 \\u279C Transform \\u279C Load to Prod \\u279C Validate\"},\"explanation\":\"Fan-out topology allows multiple independent tasks to run in parallel. In this example, after extracting data from S3, multiple tasks (Transform) are run in parallel before loading to
```

```
Prod.\",\"correct_answer\":\"A\"},{\"question\":\"In a DAG, if Task A depends on Task B, what happens
if Task A fails?\",\"options\":{\"A\":\"Task B will also fail because of the
dependency.\",\"B\":\"Task B will continue to run as Task A's failure does not affect
it.\",\"C\":\"The entire DAG will fail because of the dependency.\",\"D\":\"The failure will propagate
downstream, and Task B will wait until Task A is fixed.\"},\"explanation\":\"In a DAG, once a task is
completed, it cannot be retried unless it's rerun manually. If Task A fails, Task B will still
continue to run as Task A's failure does not affect it. However, the failure will propagate
downstream, potentially causing issues with downstream tasks if they depend on Task A's
results.\",\"correct_answer\":\"B\"},{\"question\":\"Which of the following is a correct example of a
fan-in topology in a DAG?\",\"options\":{\"A\":\"Extract from S3 \\u279C Load to Prod \\u279C
Transform\",\"B\":\"Extract from S3 \\u279C Validate \\u279C Load to Prod \\u279C
Transform\",\"C\":\"Extract from S3 \\u279C Load to Prod \\u279C Transform \\u279C
Validate\",\"D\":\"Extract from S3 \\u279C Load to Prod \\u279C Validate \\u279C
Transform\"},\"explanation\":\"Fan-in topology requires multiple upstream tasks to be completed before
a downstream task can start. Here, after Extract from S3, multiple tasks (Load to Prod and Transform)
are dependent on the completion of the extract task.\",\"correct_answer\":\"B\"},{\"question\":\"How
does acyclicity in a DAG help in preventing infinite loops?\",\"options\":{\"A\":\"By ensuring tasks
can only go in one direction, avoiding cycles.\",\"B\":\"By allowing tasks to move in any direction,
including backward.\",\"C\":\"By preventing tasks from being executed in parallel.\",\"D\":\"By
ensuring tasks can only be executed once, not multiple times.\"},\"explanation\":\"Acyclicity in DAGs
prevents loops, meaning tasks can only move in one direction (from start to end). This prevents
infinite loops where a task might keep repeating a previous state due to
cycles.\",\"correct_answer\":\"A\"}]}";
        var responseDataDto = new ResponseDataDto(goodApiResponse, 1);

        assertThrows(IllegalArgumentException.class,
            () -> {
                var jsonValidation = new JsonValidation();


jsonValidation.validateJsonObject(jsonValidation.convertStringToJsonObject(responseDataDto),
responseDataDto);
            }
        );
    }
}
```

9.95 src/test/java/capstoneProject/Lorenzo/genIz/UpdateOnlineModel↩ SettingsTest.java File Reference

```
import org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.testcontainers.service.connection.Service↩
Connection;
import org.testcontainers.containers.MySQLContainer;
import org.testcontainers.junit.jupiter.Container;
import org.testcontainers.junit.jupiter.Testcontainers;
import capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModel↩
Dto;
import jakarta.transaction.Transactional;
```

Classes

- class capstoneProject.Lorenzo.genIz.UpdateOnlineModelSettingsTest

Packages

- package capstoneProject.Lorenzo.genIz

9.96 UpdateOnlineModelSettingsTest.java

Go to the documentation of this file.

```
package capstoneProject.Lorenzo.genIz;

import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.testcontainers.service.connection.ServiceConnection;
import org.testcontainers.containers.MySQLContainer;
import org.testcontainers.junit.jupiter.Container;
import org.testcontainers.junit.jupiter.Testcontainers;

import capstoneProject.Lorenzo.genIz.DAO.QuizDaoImpl;
import capstoneProject.Lorenzo.genIz.DTO.request_dto.PostReqOnlineModelDto;
import capstoneProject.Lorenzo.genIz.DTO.response_dto.ResponseOnlineModelDto;
import jakarta.transaction.Transactional;

@SpringBootTest
@Testcontainers
@Transactional
public class UpdateOnlineModelSettingsTest {

    //create the db container to test db interaction for online model settings
    @Container
    @ServiceConnection
    static MySQLContainer<?> mysql = new MySQLContainer<>("mysql:8.4.0");

    @Autowired
    private QuizDaoImpl quizDaoImpl;

    //test to verify that the value getsacutally created
    @Test
    void testSingletonRowPeristance(){
        //create the row
        PostReqOnlineModelDto postReqOnlineModelDto = new PostReqOnlineModelDto();
        postReqOnlineModelDto.setModel_url("https://test-url/v1/completions");
        postReqOnlineModelDto.setModel_name("qwen2.5-3b");
        postReqOnlineModelDto.setApi_key("apiKeyValue1234");
        quizDaoImpl.saveNewOnlineModelSettings(postReqOnlineModelDto);

        //fetch api key back to test if data gets saved correclty
        ResponseOnlineModelDto responseOnlineModelDto = quizDaoImpl.retrieveOnlineModelSettings();
        assertEquals("apiKeyValue1234", responseOnlineModelDto.getApi_key());
    }

    //test to verify that delete command delete all the values in the row without removing the row itself
    @Test
    void testDeleteAllModelInfo(){
        quizDaoImpl.deleteOnlineModelSettings();

        //fetch api key back to test if data gets removed correclty
        ResponseOnlineModelDto responseOnlineModelDto = quizDaoImpl.retrieveOnlineModelSettings();
        assertEquals("null", responseOnlineModelDto.getApi_key());
    }

    //test to verify that each field can be modified individually
    @Test
    void testInsertNewApiKeyAndVerifyThatOtherFieldsRemainNull(){
        PostReqOnlineModelDto postReqOnlineModelDto = new PostReqOnlineModelDto();
        postReqOnlineModelDto.setApi_key("apiKey938475");

        //fetch api key back to test if data gets saved correclty
        ResponseOnlineModelDto responseOnlineModelDto = quizDaoImpl.retrieveOnlineModelSettings();
        assertEquals("apiKey938475", responseOnlineModelDto.getApi_key());
    }

    @Test
    void verifyThatOtherFieldsDidNotChanged(){
        //fetch model name to test if data gets handled correclty
        ResponseOnlineModelDto responseOnlineModelDto = quizDaoImpl.retrieveOnlineModelSettings();
        assertEquals("null", responseOnlineModelDto.getModel_name());
    }
}
```